Summary Tables of `MAXCUT` Data

1 Tables for depth $P = 1$

	ER	HH	LB	RR
		MPS (Aer)		
Fixed Angles*	10 [50]	-	9 [100]	7 [100]
Fixed Angles†	10 [50]	-	9 [100]	7 [100]
Fourier*	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
Interp.*	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
TQA*	10 [50]	10 [144]	10 [100]	7 [100]
TQA	10 [50]	10 [144]	10 [100]	7 [100]
		MPS (Quimb)		
Fixed Angles*	10 [50]	-	9 [100]	7 [100]
Fixed Angles†	10 [50]	-	9 [100]	7 [100]
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–100]
Interp.*	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS	20 [40–50]	31 [39–144]	10 [100]	18 [40–100]
TQA*	10 [50]	10 [144]	10 [100]	7 [100]
TQA	10 [50]	10 [144]	10 [100]	7 [100]
		PP		
Fixed Angles*	10 [50]	-	9 [100]	7 [100]
Fixed Angles†	10 [50]	-	9 [100]	7 [100]
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–100]
Interp.*	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–100]
Linear Ramp*	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp†	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Recursive TS	20 [40–50]	31 [39–144]	111 [40–100]	18 [40–100]
TQA*	10 [50]	10 [144]	10 [100]	7 [100]
TQA	10 [50]	10 [144]	10 [100]	7 [100]
		SV		
Fixed Angles*	64 [10–20]	-	19 [20]	273 [10–20]
Fixed Angles†	64 [10–20]	-	19 [20]	289 [10–20]
Fourier*	92 [10–20]	20 [12–21]	784 [10–20]	570 [10–20]
Interp.*	92 [10–20]	20 [12–21]	828 [10–22]	382 [10–20]
Linear Ramp*	-	-	-	-
Linear Ramp	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp†	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS	92 [10–20]	20 [12–21]	505 [10–22]	381 [10–20]
TQA*	72 [10–20]	10 [21]	603 [10–20]	66 [10–20]
TQA	74 [10–20]	10 [21]	611 [10–20]	280 [10–20]

Table 1: **Number of Instances (num. nodes in brackets) for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angles*	76.0 ± 1.2	-	72.4 ± 2.3	71.3 ± 1.3
Fixed Angles†	75.4 ± 1.3	-	72.2 ± 2.3	70.5 ± 1.6
Fourier*	81.5 ± 1.2	75.6 ± 1.2	73.1 ± 2.1	77.3 ± 2.1
Interp.*	81.5 ± 1.2	75.6 ± 1.2	73.1 ± 2.1	77.3 ± 2.1
Linear Ramp*	75.8 ± 1.3	74.6 ± 0.9	71.9 ± 2.7	71.2 ± 1.3
Linear Ramp	75.7 ± 1.3	74.6 ± 0.9	71.9 ± 2.7	71.2 ± 1.3
Linear Ramp†	73.7 ± 1.3	70.7 ± 0.8	68.2 ± 3.4	69.6 ± 2.0
Recursive TS	79.7 ± 1.2	74.8 ± 1.2	71.2 ± 2.4	75.2 ± 1.3
TQA*	75.6 ± 1.3	74.7 ± 1.0	71.8 ± 2.7	71.2 ± 1.4
TQA	75.5 ± 1.3	74.5 ± 1.0	68.9 ± 4.3	71.0 ± 1.4
MPS (Quimb)				
Fixed Angles*	75.6 ± 1.2	-	72.3 ± 2.2	69.3 ± 2.6
Fixed Angles†	74.9 ± 1.1	-	72.3 ± 2.2	68.1 ± 2.9
Fourier*	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	77.1 ± 1.8
Interp.*	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	77.1 ± 1.8
Linear Ramp*	75.8 ± 1.1	67.4 ± 1.3	71.9 ± 2.6	69.3 ± 2.5
Linear Ramp	75.8 ± 1.1	68.2 ± 1.1	71.9 ± 2.6	69.3 ± 2.6
Linear Ramp†	73.5 ± 1.1	65.8 ± 0.9	68.0 ± 3.6	67.5 ± 3.3
Recursive TS	79.7 ± 1.2	74.6 ± 1.1	71.3 ± 2.0	76.3 ± 2.0
TQA*	75.8 ± 1.1	67.4 ± 1.3	71.9 ± 2.6	69.4 ± 2.4
TQA	75.8 ± 1.2	67.3 ± 1.2	68.6 ± 4.5	68.9 ± 3.0
PP				
Fixed Angles*	81.9 ± 1.1	-	73.0 ± 1.4	78.8 ± 2.2
Fixed Angles†	81.6 ± 1.0	-	73.0 ± 1.4	78.7 ± 2.2
Fourier*	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	77.1 ± 1.8
Interp.*	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	77.1 ± 1.8
Linear Ramp*	81.9 ± 1.1	75.3 ± 1.0	72.6 ± 1.6	78.4 ± 2.3
Linear Ramp	81.9 ± 1.1	75.3 ± 1.0	72.6 ± 1.6	78.4 ± 2.3
Linear Ramp†	77.8 ± 0.9	71.3 ± 0.9	68.9 ± 3.3	75.7 ± 2.0
Recursive TS	79.7 ± 1.2	74.6 ± 1.1	71.3 ± 2.0	76.3 ± 2.0
TQA*	81.9 ± 1.1	75.4 ± 0.9	72.9 ± 1.4	78.8 ± 2.2
TQA	77.8 ± 0.9	75.1 ± 0.9	68.9 ± 4.2	76.9 ± 1.0
SV				
Fixed Angles*	79.0 ± 2.8	-	73.4 ± 2.6	81.7 ± 4.3
Fixed Angles†	78.4 ± 2.5	-	73.4 ± 2.6	80.7 ± 3.1
Fourier*	78.8 ± 2.8	78.3 ± 2.6	74.7 ± 4.0	82.0 ± 4.1
Interp.*	78.8 ± 2.8	78.3 ± 2.6	74.6 ± 3.9	81.9 ± 4.2
Linear Ramp*	-	-	-	-
Linear Ramp	78.4 ± 1.9	77.4 ± 1.7	74.2 ± 1.9	79.9 ± 2.8
Linear Ramp†	75.6 ± 1.7	71.6 ± 2.9	69.3 ± 3.1	75.7 ± 2.1
Recursive TS	78.8 ± 2.8	78.3 ± 2.6	74.9 ± 4.8	81.9 ± 4.2
TQA*	78.7 ± 2.9	77.4 ± 1.7	74.7 ± 3.6	78.3 ± 2.2
TQA	77.6 ± 2.5	75.9 ± 2.7	71.0 ± 5.0	79.0 ± 2.9

Table 2: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for MAXCUT for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	RR
MPS (Aer)								
Fixed Angles*	76.0 ± 1.2	-	72.4 ± 2.3	71.3 ± 1.3	10 [50]	-	9 [100]	7 [100]
Fixed Angles†	75.4 ± 1.3	-	72.2 ± 2.3	70.5 ± 1.6	10 [50]	-	9 [100]	7 [100]
Fourier*	81.5 ± 1.2	75.6 ± 1.2	73.1 ± 2.1	77.3 ± 2.1	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–10
Interp.*	81.5 ± 1.2	75.6 ± 1.2	73.1 ± 2.1	77.3 ± 2.1	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–10
Linear Ramp*	75.8 ± 1.3	74.6 ± 0.9	71.9 ± 2.7	71.2 ± 1.3	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	75.7 ± 1.3	74.6 ± 0.9	71.9 ± 2.7	71.2 ± 1.3	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	73.7 ± 1.3	70.7 ± 0.8	68.2 ± 3.4	69.6 ± 2.0	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS	79.7 ± 1.2	74.8 ± 1.2	71.2 ± 2.4	75.2 ± 1.3	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–10
TQA*	75.6 ± 1.3	74.7 ± 1.0	71.8 ± 2.7	71.2 ± 1.4	10 [50]	10 [144]	10 [100]	7 [100]
TQA	75.5 ± 1.3	74.5 ± 1.0	68.9 ± 4.3	71.0 ± 1.4	10 [50]	10 [144]	10 [100]	7 [100]
MPS (Quimb)								
Fixed Angles*	75.6 ± 1.2	-	72.3 ± 2.2	69.3 ± 2.6	10 [50]	-	9 [100]	7 [100]
Fixed Angles†	74.9 ± 1.1	-	72.3 ± 2.2	68.1 ± 2.9	10 [50]	-	9 [100]	7 [100]
Fourier*	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	77.1 ± 1.8	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–10
Interp.*	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	77.1 ± 1.8	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–10
Linear Ramp*	75.8 ± 1.1	67.4 ± 1.3	71.9 ± 2.6	69.3 ± 2.5	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	75.8 ± 1.1	68.2 ± 1.1	71.9 ± 2.6	69.3 ± 2.6	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	73.5 ± 1.1	65.8 ± 0.9	68.0 ± 3.6	67.5 ± 3.3	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS	79.7 ± 1.2	74.6 ± 1.1	71.3 ± 2.0	76.3 ± 2.0	20 [40–50]	31 [39–144]	10 [100]	18 [40–10
TQA*	75.8 ± 1.1	67.4 ± 1.3	71.9 ± 2.6	69.4 ± 2.4	10 [50]	10 [144]	10 [100]	7 [100]
TQA	75.8 ± 1.2	67.3 ± 1.2	68.6 ± 4.5	68.9 ± 3.0	10 [50]	10 [144]	10 [100]	7 [100]
PP								
Fixed Angles*	81.9 ± 1.1	-	73.0 ± 1.4	78.8 ± 2.2	10 [50]	-	9 [100]	7 [100]
Fixed Angles†	81.6 ± 1.0	-	73.0 ± 1.4	78.7 ± 2.2	10 [50]	-	9 [100]	7 [100]
Fourier*	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	77.1 ± 1.8	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–10
Interp.*	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	77.1 ± 1.8	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–10
Linear Ramp*	81.9 ± 1.1	75.3 ± 1.0	72.6 ± 1.6	78.4 ± 2.3	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp	81.9 ± 1.1	75.3 ± 1.0	72.6 ± 1.6	78.4 ± 2.3	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp†	77.8 ± 0.9	71.3 ± 0.9	68.9 ± 3.3	75.7 ± 2.0	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Recursive TS	79.7 ± 1.2	74.6 ± 1.1	71.3 ± 2.0	76.3 ± 2.0	20 [40–50]	31 [39–144]	111 [40–100]	18 [40–10
TQA*	81.9 ± 1.1	75.4 ± 0.9	72.9 ± 1.4	78.8 ± 2.2	10 [50]	10 [144]	10 [100]	7 [100]
TQA	77.8 ± 0.9	75.1 ± 0.9	68.9 ± 4.2	76.9 ± 1.0	10 [50]	10 [144]	10 [100]	7 [100]
SV								
Fixed Angles*	79.0 ± 2.8	-	73.4 ± 2.6	81.7 ± 4.3	64 [10–20]	-	19 [20]	273 [10–2
Fixed Angles†	78.4 ± 2.5	-	73.4 ± 2.6	80.7 ± 3.1	64 [10–20]	-	19 [20]	289 [10–2
Fourier*	78.8 ± 2.8	78.3 ± 2.6	74.7 ± 4.0	82.0 ± 4.1	92 [10–20]	20 [12–21]	784 [10–20]	570 [10–2
Interp.*	78.8 ± 2.8	78.3 ± 2.6	74.6 ± 3.9	81.9 ± 4.2	92 [10–20]	20 [12–21]	828 [10–22]	382 [10–2
Linear Ramp*	-	-	-	-	-	-	-	-
Linear Ramp	78.4 ± 1.9	77.4 ± 1.7	74.2 ± 1.9	79.9 ± 2.8	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp†	75.6 ± 1.7	71.6 ± 2.9	69.3 ± 3.1	75.7 ± 2.1	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS	78.8 ± 2.8	78.3 ± 2.6	74.9 ± 4.8	81.9 ± 4.2	92 [10–20]	20 [12–21]	505 [10–22]	381 [10–2
TQA*	78.7 ± 2.9	77.4 ± 1.7	74.7 ± 3.6	78.3 ± 2.2	72 [10–20]	10 [21]	603 [10–20]	66 [10–20
TQA	77.6 ± 2.5	75.9 ± 2.7	71.0 ± 5.0	79.0 ± 2.9	74 [10–20]	10 [21]	611 [10–20]	280 [10–2

Table 3: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

2 Tables for depth $P = 2$

	ER	HH	LB	RR
		MPS (Aer)		
Fixed Angles*	10 [50]	10 [39]	27 [100]	17 [40–100]
Fixed Angles†	10 [50]	10 [39]	27 [100]	17 [40–100]
Fourier*	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
Interp.*	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	19 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	19 [100]	7 [100]
Recursive TS	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
TQA*	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
TQA	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
		MPS (Quimb)		
Fixed Angles*	10 [50]	10 [144]	90 [100]	47 [30–100]
Fixed Angles†	10 [50]	10 [144]	90 [100]	47 [30–100]
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–100]
Interp.*	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS	20 [40–50]	31 [39–144]	10 [100]	18 [40–100]
TQA*	20 [40–50]	20 [39–144]	130 [40–100]	57 [30–100]
TQA	20 [40–50]	20 [39–144]	130 [40–100]	57 [30–100]
		PP		
Fixed Angles*	10 [50]	30 [39–144]	82 [100]	47 [30–100]
Fixed Angles†	10 [50]	30 [39–144]	82 [100]	47 [30–100]
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–100]
Interp.*	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–100]
Linear Ramp*	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp	10 [50]	20 [105–144]	29 [40–100]	8 [100]
Linear Ramp†	10 [50]	20 [105–144]	29 [40–100]	8 [100]
Recursive TS	20 [40–50]	31 [39–144]	111 [40–100]	18 [40–100]
TQA*	20 [40–50]	20 [39–144]	130 [40–100]	57 [30–100]
TQA	20 [40–50]	20 [39–144]	130 [40–100]	57 [30–100]
		SV		
Fixed Angles*	81 [10–20]	-	311 [10–22]	360 [10–20]
Fixed Angles†	81 [10–20]	-	311 [10–22]	372 [10–20]
Fourier*	92 [10–20]	20 [12–21]	784 [10–20]	567 [10–20]
Interp.*	92 [10–20]	20 [12–21]	828 [10–22]	382 [10–20]
Linear Ramp*	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp†	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS	92 [10–20]	20 [12–21]	505 [10–22]	381 [10–20]
TQA*	92 [10–20]	20 [12–21]	821 [10–22]	142 [10–20]
TQA	92 [10–20]	20 [12–21]	822 [10–22]	361 [10–20]

Table 4: **Number of Instances (num. nodes in brackets) for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
		MPS (Aer)		
Fixed Angles*	77.0 ± 1.9	82.2 ± 1.3	75.3 ± 3.9	75.6 ± 4.9
Fixed Angles†	75.6 ± 1.3	80.9 ± 1.5	70.6 ± 6.4	74.7 ± 5.1
Fourier*	71.5 ± 2.3	75.6 ± 3.8	69.4 ± 4.3	73.1 ± 3.5
Interp.*	77.0 ± 1.6	80.4 ± 2.0	73.1 ± 5.2	70.9 ± 2.2
Linear Ramp*	77.9 ± 2.1	77.9 ± 1.2	78.4 ± 4.3	70.1 ± 2.2
Linear Ramp	76.3 ± 1.2	77.9 ± 1.6	73.0 ± 5.9	70.1 ± 2.2
Linear Ramp†	75.6 ± 1.2	72.0 ± 1.2	70.4 ± 5.7	68.9 ± 2.8
Recursive TS	76.6 ± 1.5	77.8 ± 2.9	75.2 ± 7.0	73.9 ± 2.9
TQA*	76.0 ± 1.2	78.8 ± 2.9	72.2 ± 6.2	74.4 ± 4.6
TQA	69.0 ± 1.7	72.7 ± 1.4	67.1 ± 3.9	72.1 ± 2.8
		MPS (Quimb)		
Fixed Angles*	80.5 ± 1.9	53.9 ± 0.6	79.9 ± 2.4	76.4 ± 4.9
Fixed Angles†	77.2 ± 0.9	52.5 ± 0.6	79.6 ± 2.6	75.3 ± 5.5
Fourier*	70.6 ± 1.8	72.1 ± 8.3	76.1 ± 4.3	74.6 ± 4.7
Interp.*	79.7 ± 2.0	75.9 ± 6.2	79.2 ± 3.0	74.9 ± 4.4
Linear Ramp*	79.3 ± 2.5	68.7 ± 1.4	79.6 ± 3.0	70.4 ± 5.3
Linear Ramp	78.5 ± 3.2	69.5 ± 1.3	79.0 ± 3.3	70.3 ± 4.0
Linear Ramp†	75.7 ± 0.9	66.2 ± 0.9	77.1 ± 2.5	67.6 ± 3.5
Recursive TS	78.9 ± 2.7	66.6 ± 10.7	78.2 ± 3.3	70.6 ± 4.6
TQA*	77.4 ± 1.3	68.6 ± 14.9	79.2 ± 3.0	76.5 ± 4.8
TQA	69.3 ± 1.1	63.5 ± 9.8	70.0 ± 3.9	72.2 ± 3.0
		PP		
Fixed Angles*	87.8 ± 1.0	82.7 ± 1.2	79.2 ± 1.6	83.8 ± 1.5
Fixed Angles†	86.9 ± 1.0	81.2 ± 1.4	79.2 ± 1.6	83.7 ± 1.5
Fourier*	86.7 ± 2.1	80.8 ± 2.2	79.1 ± 1.7	80.5 ± 2.7
Interp.*	87.4 ± 1.2	82.7 ± 1.2	79.6 ± 2.0	83.8 ± 1.4
Linear Ramp*	87.8 ± 1.0	82.5 ± 1.0	80.3 ± 1.9	83.9 ± 1.1
Linear Ramp	86.9 ± 1.0	82.4 ± 1.0	80.5 ± 1.8	83.9 ± 1.2
Linear Ramp†	85.7 ± 0.9	73.9 ± 1.2	78.0 ± 1.2	81.6 ± 3.8
Recursive TS	85.2 ± 1.2	81.3 ± 1.2	79.2 ± 1.7	80.3 ± 2.4
TQA*	87.4 ± 1.2	82.8 ± 1.4	79.6 ± 2.0	83.8 ± 1.4
TQA	78.9 ± 1.3	72.8 ± 1.4	70.6 ± 3.0	77.1 ± 2.0
		SV		
Fixed Angles*	85.4 ± 2.2	-	82.5 ± 2.9	88.1 ± 3.3
Fixed Angles†	84.1 ± 2.1	-	82.2 ± 2.9	87.2 ± 3.0
Fourier*	76.4 ± 8.8	75.1 ± 4.5	76.2 ± 5.7	80.6 ± 8.9
Interp.*	85.3 ± 2.3	84.8 ± 1.7	82.5 ± 4.0	86.4 ± 4.7
Linear Ramp*	84.9 ± 1.6	84.8 ± 1.2	82.0 ± 1.8	86.0 ± 2.5
Linear Ramp	84.6 ± 1.7	84.5 ± 1.2	81.7 ± 1.8	85.7 ± 2.4
Linear Ramp†	82.1 ± 2.1	73.7 ± 3.9	79.8 ± 1.9	83.7 ± 3.7
Recursive TS	85.2 ± 2.3	84.3 ± 3.3	82.1 ± 4.8	86.1 ± 4.3
TQA*	85.5 ± 2.3	85.7 ± 2.1	82.8 ± 3.1	87.3 ± 3.3
TQA	76.2 ± 3.1	72.9 ± 4.2	72.5 ± 4.2	75.7 ± 4.4

Table 5: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for MAXCUT for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	RR
				MPS (Aer)				
Fixed Angles*	77.0 ± 1.9	82.2 ± 1.3	75.3 ± 3.9	75.6 ± 4.9	10 [50]	10 [39]	27 [100]	17 [40–10
Fixed Angles†	75.6 ± 1.3	80.9 ± 1.5	70.6 ± 6.4	74.7 ± 5.1	10 [50]	10 [39]	27 [100]	17 [40–10
Fourier*	71.5 ± 2.3	75.6 ± 3.8	69.4 ± 4.3	73.1 ± 3.5	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–10
Interp.*	77.0 ± 1.6	80.4 ± 2.0	73.1 ± 5.2	70.9 ± 2.2	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–10
Linear Ramp*	77.9 ± 2.1	77.9 ± 1.2	78.4 ± 4.3	70.1 ± 2.2	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	76.3 ± 1.2	77.9 ± 1.6	73.0 ± 5.9	70.1 ± 2.2	10 [50]	10 [144]	19 [100]	7 [100]
Linear Ramp†	75.6 ± 1.2	72.0 ± 1.2	70.4 ± 5.7	68.9 ± 2.8	10 [50]	10 [144]	19 [100]	7 [100]
Recursive TS	76.6 ± 1.5	77.8 ± 2.9	75.2 ± 7.0	73.9 ± 2.9	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–10
TQA*	76.0 ± 1.2	78.8 ± 2.9	72.2 ± 6.2	74.4 ± 4.6	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–10
TQA	69.0 ± 1.7	72.7 ± 1.4	67.1 ± 3.9	72.1 ± 2.8	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–10
				MPS (Quimb)				
Fixed Angles*	80.5 ± 1.9	53.9 ± 0.6	79.9 ± 2.4	76.4 ± 4.9	10 [50]	10 [144]	90 [100]	47 [30–10
Fixed Angles†	77.2 ± 0.9	52.5 ± 0.6	79.6 ± 2.6	75.3 ± 5.5	10 [50]	10 [144]	90 [100]	47 [30–10
Fourier*	70.6 ± 1.8	72.1 ± 8.3	76.1 ± 4.3	74.6 ± 4.7	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–10
Interp.*	79.7 ± 2.0	75.9 ± 6.2	79.2 ± 3.0	74.9 ± 4.4	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–10
Linear Ramp*	79.3 ± 2.5	68.7 ± 1.4	79.6 ± 3.0	70.4 ± 5.3	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	78.5 ± 3.2	69.5 ± 1.3	79.0 ± 3.3	70.3 ± 4.0	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	75.7 ± 0.9	66.2 ± 0.9	77.1 ± 2.5	67.6 ± 3.5	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS	78.9 ± 2.7	66.6 ± 10.7	78.2 ± 3.3	70.6 ± 4.6	20 [40–50]	31 [39–144]	10 [100]	18 [40–10
TQA*	77.4 ± 1.3	68.6 ± 14.9	79.2 ± 3.0	76.5 ± 4.8	20 [40–50]	20 [39–144]	130 [40–100]	57 [30–10
TQA	69.3 ± 1.1	63.5 ± 9.8	70.0 ± 3.9	72.2 ± 3.0	20 [40–50]	20 [39–144]	130 [40–100]	57 [30–10
				PP				
Fixed Angles*	87.8 ± 1.0	82.7 ± 1.2	79.2 ± 1.6	83.8 ± 1.5	10 [50]	30 [39–144]	82 [100]	47 [30–10
Fixed Angles†	86.9 ± 1.0	81.2 ± 1.4	79.2 ± 1.6	83.7 ± 1.5	10 [50]	30 [39–144]	82 [100]	47 [30–10
Fourier*	86.7 ± 2.1	80.8 ± 2.2	79.1 ± 1.7	80.5 ± 2.7	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–10
Interp.*	87.4 ± 1.2	82.7 ± 1.2	79.6 ± 2.0	83.8 ± 1.4	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–10
Linear Ramp*	87.8 ± 1.0	82.5 ± 1.0	80.3 ± 1.9	83.9 ± 1.1	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp	86.9 ± 1.0	82.4 ± 1.0	80.5 ± 1.8	83.9 ± 1.2	10 [50]	20 [105–144]	29 [40–100]	8 [100]
Linear Ramp†	85.7 ± 0.9	73.9 ± 1.2	78.0 ± 1.2	81.6 ± 3.8	10 [50]	20 [105–144]	29 [40–100]	8 [100]
Recursive TS	85.2 ± 1.2	81.3 ± 1.2	79.2 ± 1.7	80.3 ± 2.4	20 [40–50]	31 [39–144]	111 [40–100]	18 [40–10
TQA*	87.4 ± 1.2	82.8 ± 1.4	79.6 ± 2.0	83.8 ± 1.4	20 [40–50]	20 [39–144]	130 [40–100]	57 [30–10
TQA	78.9 ± 1.3	72.8 ± 1.4	70.6 ± 3.0	77.1 ± 2.0	20 [40–50]	20 [39–144]	130 [40–100]	57 [30–10
				SV				
Fixed Angles*	85.4 ± 2.2	-	82.5 ± 2.9	88.1 ± 3.3	81 [10–20]	-	311 [10–22]	360 [10–
Fixed Angles†	84.1 ± 2.1	-	82.2 ± 2.9	87.2 ± 3.0	81 [10–20]	-	311 [10–22]	372 [10–
Fourier*	76.4 ± 8.8	75.1 ± 4.5	76.2 ± 5.7	80.6 ± 8.9	92 [10–20]	20 [12–21]	784 [10–20]	567 [10–
Interp.*	85.3 ± 2.3	84.8 ± 1.7	82.5 ± 4.0	86.4 ± 4.7	92 [10–20]	20 [12–21]	828 [10–22]	382 [10–
Linear Ramp*	84.9 ± 1.6	84.8 ± 1.2	82.0 ± 1.8	86.0 ± 2.5	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp	84.6 ± 1.7	84.5 ± 1.2	81.7 ± 1.8	85.7 ± 2.4	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp†	82.1 ± 2.1	73.7 ± 3.9	79.8 ± 1.9	83.7 ± 3.7	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS	85.2 ± 2.3	84.3 ± 3.3	82.1 ± 4.8	86.1 ± 4.3	92 [10–20]	20 [12–21]	505 [10–22]	381 [10–
TQA*	85.5 ± 2.3	85.7 ± 2.1	82.8 ± 3.1	87.3 ± 3.3	92 [10–20]	20 [12–21]	821 [10–22]	142 [10–
TQA	76.2 ± 3.1	72.9 ± 4.2	72.5 ± 4.2	75.7 ± 4.4	92 [10–20]	20 [12–21]	822 [10–22]	361 [10–

Table 6: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

3 Tables for depth $P = 3$

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angles*	7 [50]	-	8 [100]	7 [100]
Fixed Angles†	7 [50]	-	8 [100]	7 [100]
Fourier*	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
Interp.*	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
TQA*	10 [50]	10 [144]	10 [100]	7 [100]
TQA	10 [50]	10 [144]	10 [100]	7 [100]
MPS (Quimb)				
Fixed Angles*	7 [50]	10 [144]	8 [100]	37 [30–100]
Fixed Angles†	7 [50]	10 [144]	8 [100]	37 [30–100]
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	48 [30–100]
Interp.*	20 [40–50]	31 [39–144]	130 [40–100]	48 [30–100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS	20 [40–50]	31 [39–144]	10 [100]	18 [40–100]
TQA*	10 [50]	10 [144]	20 [40–100]	37 [30–100]
TQA	10 [50]	10 [144]	20 [40–100]	37 [30–100]
PP				
Fixed Angles*	7 [50]	10 [144]	8 [100]	37 [30–100]
Fixed Angles†	7 [50]	10 [144]	8 [100]	37 [30–100]
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	48 [30–100]
Interp.*	20 [40–50]	31 [39–144]	130 [40–100]	48 [30–100]
Linear Ramp*	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp†	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Recursive TS	20 [40–50]	31 [39–144]	111 [40–100]	18 [40–100]
TQA*	10 [50]	10 [144]	20 [40–100]	37 [30–100]
TQA	10 [50]	10 [144]	20 [40–100]	37 [30–100]
SV				
Fixed Angles*	81 [10–20]	-	283 [10–20]	365 [10–20]
Fixed Angles†	81 [10–20]	-	283 [10–20]	372 [10–20]
Fourier*	92 [10–20]	20 [12–21]	784 [10–20]	567 [10–20]
Interp.*	92 [10–20]	20 [12–21]	827 [10–22]	382 [10–20]
Linear Ramp*	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp†	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS	92 [10–20]	20 [12–21]	505 [10–22]	381 [10–20]
TQA*	91 [10–20]	20 [12–21]	821 [10–22]	141 [10–20]
TQA	92 [10–20]	20 [12–21]	821 [10–22]	357 [10–20]

Table 7: **Number of Instances (num. nodes in brackets) for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
		MPS (Aer)		
Fixed Angles*	78.7 ± 3.5	-	87.9 ± 1.3	72.6 ± 4.6
Fixed Angles†	75.2 ± 1.4	-	84.6 ± 3.9	68.6 ± 2.3
Fourier*	79.9 ± 4.0	77.9 ± 2.7	72.0 ± 7.4	74.5 ± 4.0
Interp.*	81.1 ± 3.7	83.4 ± 3.0	80.1 ± 7.0	72.3 ± 3.3
Linear Ramp*	78.6 ± 3.6	79.3 ± 1.2	87.9 ± 1.4	70.9 ± 3.2
Linear Ramp	77.2 ± 2.1	77.5 ± 1.4	85.9 ± 1.6	70.2 ± 2.8
Linear Ramp†	75.2 ± 1.2	71.6 ± 1.2	76.2 ± 4.9	67.4 ± 3.2
Recursive TS	79.4 ± 3.2	79.5 ± 3.5	81.4 ± 6.2	75.6 ± 3.7
TQA*	75.9 ± 1.2	78.6 ± 2.5	82.9 ± 4.7	70.9 ± 3.3
TQA	69.4 ± 3.9	76.7 ± 2.0	78.5 ± 5.8	69.8 ± 2.5
		MPS (Quimb)		
Fixed Angles*	84.5 ± 2.2	56.5 ± 1.5	86.4 ± 1.7	81.9 ± 6.0
Fixed Angles†	79.0 ± 2.1	54.1 ± 0.8	85.7 ± 2.3	79.1 ± 7.8
Fourier*	84.2 ± 6.6	75.6 ± 6.8	80.1 ± 4.4	77.0 ± 5.6
Interp.*	87.9 ± 1.5	79.0 ± 7.0	85.7 ± 1.6	80.5 ± 6.9
Linear Ramp*	88.7 ± 2.5	70.7 ± 1.2	86.9 ± 1.4	76.3 ± 5.1
Linear Ramp	86.0 ± 2.0	71.4 ± 1.2	84.1 ± 1.4	75.4 ± 5.4
Linear Ramp†	78.5 ± 1.7	66.5 ± 0.8	80.1 ± 1.6	66.6 ± 3.8
Recursive TS	87.6 ± 3.2	68.5 ± 11.3	83.8 ± 1.4	73.6 ± 5.6
TQA*	81.3 ± 5.0	57.5 ± 1.0	86.4 ± 3.7	79.5 ± 7.3
TQA	75.3 ± 3.6	57.0 ± 0.9	80.5 ± 3.3	74.7 ± 4.7
		PP		
Fixed Angles*	90.8 ± 1.2	86.4 ± 1.0	83.8 ± 1.6	87.4 ± 1.4
Fixed Angles†	89.7 ± 1.2	83.7 ± 1.5	83.0 ± 1.2	87.4 ± 1.3
Fourier*	86.7 ± 1.9	83.9 ± 2.7	81.0 ± 2.6	84.7 ± 2.1
Interp.*	89.0 ± 1.7	86.7 ± 1.1	83.3 ± 1.9	87.2 ± 1.4
Linear Ramp*	91.2 ± 1.1	86.5 ± 0.9	84.2 ± 1.9	86.8 ± 0.9
Linear Ramp	89.6 ± 1.0	85.4 ± 0.8	82.7 ± 1.8	85.8 ± 1.2
Linear Ramp†	89.3 ± 1.0	75.5 ± 1.3	80.5 ± 1.1	83.3 ± 3.7
Recursive TS	86.7 ± 1.2	83.6 ± 0.9	82.6 ± 1.5	82.8 ± 1.7
TQA*	89.4 ± 1.4	83.5 ± 0.8	80.0 ± 2.1	86.9 ± 1.6
TQA	80.7 ± 0.8	79.1 ± 1.0	71.4 ± 4.6	81.0 ± 1.5
		SV		
Fixed Angles*	89.1 ± 1.8	-	87.7 ± 2.6	91.6 ± 2.8
Fixed Angles†	87.1 ± 1.7	-	87.2 ± 2.5	89.9 ± 2.2
Fourier*	83.8 ± 5.6	79.8 ± 5.3	78.7 ± 7.5	88.5 ± 4.0
Interp.*	88.8 ± 1.9	89.0 ± 1.6	87.4 ± 3.8	89.6 ± 4.1
Linear Ramp*	88.5 ± 1.4	88.8 ± 1.3	86.9 ± 1.9	89.5 ± 2.0
Linear Ramp	87.7 ± 1.5	86.8 ± 1.8	84.9 ± 1.8	88.5 ± 2.3
Linear Ramp†	83.9 ± 2.0	75.1 ± 4.1	82.6 ± 2.3	85.9 ± 4.2
Recursive TS	87.8 ± 2.4	87.3 ± 3.3	85.7 ± 4.9	88.5 ± 3.8
TQA*	87.7 ± 2.2	87.0 ± 2.1	85.1 ± 3.8	89.8 ± 2.6
TQA	81.3 ± 3.8	79.5 ± 4.1	77.3 ± 5.1	80.1 ± 5.5

Table 8: **Avg. Approx. Ratio ± standard deviation in percentage for MAXCUT for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	RR
MPS (Aer)								
Fixed Angles*	78.7 ± 3.5	-	87.9 ± 1.3	72.6 ± 4.6	7 [50]	-	8 [100]	7 [100]
Fixed Angles†	75.2 ± 1.4	-	84.6 ± 3.9	68.6 ± 2.3	7 [50]	-	8 [100]	7 [100]
Fourier*	79.9 ± 4.0	77.9 ± 2.7	72.0 ± 7.4	74.5 ± 4.0	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–10
Interp.*	81.1 ± 3.7	83.4 ± 3.0	80.1 ± 7.0	72.3 ± 3.3	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–10
Linear Ramp*	78.6 ± 3.6	79.3 ± 1.2	87.9 ± 1.4	70.9 ± 3.2	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	77.2 ± 2.1	77.5 ± 1.4	85.9 ± 1.6	70.2 ± 2.8	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	75.2 ± 1.2	71.6 ± 1.2	76.2 ± 4.9	67.4 ± 3.2	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS	79.4 ± 3.2	79.5 ± 3.5	81.4 ± 6.2	75.6 ± 3.7	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–10
TQA*	75.9 ± 1.2	78.6 ± 2.5	82.9 ± 4.7	70.9 ± 3.3	10 [50]	10 [144]	10 [100]	7 [100]
TQA	69.4 ± 3.9	76.7 ± 2.0	78.5 ± 5.8	69.8 ± 2.5	10 [50]	10 [144]	10 [100]	7 [100]
MPS (Quimb)								
Fixed Angles*	84.5 ± 2.2	56.5 ± 1.5	86.4 ± 1.7	81.9 ± 6.0	7 [50]	10 [144]	8 [100]	37 [30–10
Fixed Angles†	79.0 ± 2.1	54.1 ± 0.8	85.7 ± 2.3	79.1 ± 7.8	7 [50]	10 [144]	8 [100]	37 [30–10
Fourier*	84.2 ± 6.6	75.6 ± 6.8	80.1 ± 4.4	77.0 ± 5.6	20 [40–50]	31 [39–144]	130 [40–100]	48 [30–10
Interp.*	87.9 ± 1.5	79.0 ± 7.0	85.7 ± 1.6	80.5 ± 6.9	20 [40–50]	31 [39–144]	130 [40–100]	48 [30–10
Linear Ramp*	88.7 ± 2.5	70.7 ± 1.2	86.9 ± 1.4	76.3 ± 5.1	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	86.0 ± 2.0	71.4 ± 1.2	84.1 ± 1.4	75.4 ± 5.4	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	78.5 ± 1.7	66.5 ± 0.8	80.1 ± 1.6	66.6 ± 3.8	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS	87.6 ± 3.2	68.5 ± 11.3	83.8 ± 1.4	73.6 ± 5.6	20 [40–50]	31 [39–144]	10 [100]	18 [40–10
TQA*	81.3 ± 5.0	57.5 ± 1.0	86.4 ± 3.7	79.5 ± 7.3	10 [50]	10 [144]	20 [40–100]	37 [30–10
TQA	75.3 ± 3.6	57.0 ± 0.9	80.5 ± 3.3	74.7 ± 4.7	10 [50]	10 [144]	20 [40–100]	37 [30–10
PP								
Fixed Angles*	90.8 ± 1.2	86.4 ± 1.0	83.8 ± 1.6	87.4 ± 1.4	7 [50]	10 [144]	8 [100]	37 [30–10
Fixed Angles†	89.7 ± 1.2	83.7 ± 1.5	83.0 ± 1.2	87.4 ± 1.3	7 [50]	10 [144]	8 [100]	37 [30–10
Fourier*	86.7 ± 1.9	83.9 ± 2.7	81.0 ± 2.6	84.7 ± 2.1	20 [40–50]	31 [39–144]	130 [40–100]	48 [30–10
Interp.*	89.0 ± 1.7	86.7 ± 1.1	83.3 ± 1.9	87.2 ± 1.4	20 [40–50]	31 [39–144]	130 [40–100]	48 [30–10
Linear Ramp*	91.2 ± 1.1	86.5 ± 0.9	84.2 ± 1.9	86.8 ± 0.9	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp	89.6 ± 1.0	85.4 ± 0.8	82.7 ± 1.8	85.8 ± 1.2	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp†	89.3 ± 1.0	75.5 ± 1.3	80.5 ± 1.1	83.3 ± 3.7	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Recursive TS	86.7 ± 1.2	83.6 ± 0.9	82.6 ± 1.5	82.8 ± 1.7	20 [40–50]	31 [39–144]	111 [40–100]	18 [40–10
TQA*	89.4 ± 1.4	83.5 ± 0.8	80.0 ± 2.1	86.9 ± 1.6	10 [50]	10 [144]	20 [40–100]	37 [30–10
TQA	80.7 ± 0.8	79.1 ± 1.0	71.4 ± 4.6	81.0 ± 1.5	10 [50]	10 [144]	20 [40–100]	37 [30–10
SV								
Fixed Angles*	89.1 ± 1.8	-	87.7 ± 2.6	91.6 ± 2.8	81 [10–20]	-	283 [10–20]	365 [10–2
Fixed Angles†	87.1 ± 1.7	-	87.2 ± 2.5	89.9 ± 2.2	81 [10–20]	-	283 [10–20]	372 [10–2
Fourier*	83.8 ± 5.6	79.8 ± 5.3	78.7 ± 7.5	88.5 ± 4.0	92 [10–20]	20 [12–21]	784 [10–20]	567 [10–2
Interp.*	88.8 ± 1.9	89.0 ± 1.6	87.4 ± 3.8	89.6 ± 4.1	92 [10–20]	20 [12–21]	827 [10–22]	382 [10–2
Linear Ramp*	88.5 ± 1.4	88.8 ± 1.3	86.9 ± 1.9	89.5 ± 2.0	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp	87.7 ± 1.5	86.8 ± 1.8	84.9 ± 1.8	88.5 ± 2.3	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp†	83.9 ± 2.0	75.1 ± 4.1	82.6 ± 2.3	85.9 ± 4.2	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS	87.8 ± 2.4	87.3 ± 3.3	85.7 ± 4.9	88.5 ± 3.8	92 [10–20]	20 [12–21]	505 [10–22]	381 [10–2
TQA*	87.7 ± 2.2	87.0 ± 2.1	85.1 ± 3.8	89.8 ± 2.6	91 [10–20]	20 [12–21]	821 [10–22]	141 [10–2
TQA	81.3 ± 3.8	79.5 ± 4.1	77.3 ± 5.1	80.1 ± 5.5	92 [10–20]	20 [12–21]	821 [10–22]	357 [10–2

Table 9: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

4 Tables for depth $P = 4$

	ER	HH	LB	RR
		MPS (Aer)		
Fixed Angles*	-	10 [39]	21 [100]	13 [40–100]
Fixed Angles†	-	10 [39]	21 [100]	13 [40–100]
Fourier*	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
Interp.*	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
TQA*	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
TQA	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
		MPS (Quimb)		
Fixed Angles*	-	10 [144]	30 [100]	13 [40–100]
Fixed Angles†	-	10 [144]	30 [100]	13 [40–100]
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	28 [40–100]
Interp.*	20 [40–50]	31 [39–144]	130 [40–100]	28 [40–100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS	20 [40–50]	31 [39–144]	10 [100]	18 [40–100]
TQA*	20 [40–50]	20 [39–144]	130 [40–100]	27 [40–100]
TQA	20 [40–50]	20 [39–144]	130 [40–100]	27 [40–100]
		PP		
Fixed Angles*	-	30 [39–144]	21 [100]	13 [40–100]
Fixed Angles†	-	30 [39–144]	21 [100]	13 [40–100]
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	28 [40–100]
Interp.*	20 [40–50]	31 [39–144]	130 [40–100]	28 [40–100]
Linear Ramp*	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp†	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Recursive TS	20 [40–50]	31 [39–144]	111 [40–100]	18 [40–100]
TQA*	20 [40–50]	20 [39–144]	130 [40–100]	27 [40–100]
TQA	20 [40–50]	20 [39–144]	130 [40–100]	27 [40–100]
		SV		
Fixed Angles*	61 [10–20]	-	106 [10–20]	155 [10–20]
Fixed Angles†	61 [10–20]	-	106 [10–20]	158 [10–20]
Fourier*	92 [10–20]	20 [12–21]	784 [10–20]	567 [10–20]
Interp.*	92 [10–20]	20 [12–21]	827 [10–22]	382 [10–20]
Linear Ramp*	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp†	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS	92 [10–20]	20 [12–21]	504 [10–22]	381 [10–20]
TQA*	91 [10–20]	20 [12–21]	821 [10–22]	142 [10–20]
TQA	92 [10–20]	20 [12–21]	821 [10–22]	355 [10–20]

Table 10: **Number of Instances (num. nodes in brackets) for** $P = 4$**.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angles*	-	88.8 ± 1.3	82.8 ± 6.9	88.6 ± 8.8
Fixed Angles†	-	87.9 ± 1.5	77.1 ± 8.3	85.4 ± 11.0
Fourier*	75.6 ± 4.5	80.9 ± 2.7	74.1 ± 8.5	76.6 ± 5.5
Interp.*	87.0 ± 2.6	85.7 ± 3.4	86.2 ± 6.2	77.7 ± 6.7
Linear Ramp*	85.8 ± 4.1	81.4 ± 1.0	91.1 ± 1.2	75.0 ± 6.0
Linear Ramp	81.7 ± 4.7	79.9 ± 1.7	88.8 ± 1.8	74.5 ± 5.8
Linear Ramp†	75.4 ± 1.5	73.4 ± 1.5	82.5 ± 1.1	66.5 ± 3.4
Recursive TS	84.7 ± 4.2	81.0 ± 4.1	87.7 ± 3.7	78.8 ± 4.9
TQA*	77.8 ± 4.9	82.5 ± 2.7	78.7 ± 9.6	79.2 ± 6.3
TQA	69.6 ± 1.8	79.8 ± 2.1	74.4 ± 9.2	75.4 ± 5.1
MPS (Quimb)				
Fixed Angles*	-	52.8 ± 1.1	90.5 ± 1.4	89.5 ± 5.9
Fixed Angles†	-	51.0 ± 0.9	90.4 ± 1.5	84.8 ± 10.7
Fourier*	77.4 ± 6.4	76.0 ± 8.2	83.0 ± 4.4	83.6 ± 9.6
Interp.*	92.1 ± 1.1	79.5 ± 8.6	89.9 ± 1.1	87.3 ± 8.4
Linear Ramp*	92.4 ± 1.9	68.8 ± 2.0	90.2 ± 1.0	77.8 ± 7.4
Linear Ramp	90.7 ± 1.5	69.4 ± 2.0	87.7 ± 1.6	76.7 ± 5.8
Linear Ramp†	83.6 ± 2.6	64.4 ± 1.2	83.4 ± 1.4	66.0 ± 4.1
Recursive TS	92.6 ± 1.6	68.8 ± 12.3	88.7 ± 1.9	78.5 ± 7.4
TQA*	79.0 ± 7.0	70.7 ± 16.6	88.5 ± 1.9	84.2 ± 7.9
TQA	70.5 ± 3.4	68.1 ± 14.3	84.2 ± 2.4	77.4 ± 5.5
PP				
Fixed Angles*	-	89.3 ± 1.1	86.2 ± 1.7	89.6 ± 1.5
Fixed Angles†	-	88.2 ± 1.5	84.2 ± 1.4	89.4 ± 1.6
Fourier*	88.0 ± 2.1	86.4 ± 2.7	82.4 ± 2.9	87.2 ± 1.8
Interp.*	90.7 ± 1.9	89.3 ± 1.1	85.3 ± 1.8	89.1 ± 1.5
Linear Ramp*	92.9 ± 1.1	88.9 ± 0.8	86.6 ± 2.0	88.6 ± 0.7
Linear Ramp	91.4 ± 1.0	87.5 ± 0.7	85.2 ± 2.3	87.9 ± 0.7
Linear Ramp†	91.2 ± 1.0	77.1 ± 1.3	82.2 ± 1.0	84.5 ± 3.5
Recursive TS	88.4 ± 1.2	85.4 ± 1.2	84.3 ± 1.8	85.4 ± 1.7
TQA*	89.2 ± 1.9	86.8 ± 1.2	83.5 ± 1.9	87.8 ± 1.4
TQA	83.7 ± 1.4	81.9 ± 1.3	75.7 ± 4.8	85.1 ± 1.8
SV				
Fixed Angles*	91.4 ± 1.6	-	91.7 ± 2.4	93.4 ± 2.2
Fixed Angles†	89.0 ± 2.1	-	90.9 ± 2.4	92.8 ± 2.2
Fourier*	78.4 ± 10.6	81.3 ± 6.1	80.7 ± 7.8	90.3 ± 4.4
Interp.*	91.2 ± 1.6	91.3 ± 1.4	90.3 ± 3.8	90.9 ± 4.9
Linear Ramp*	90.9 ± 1.2	90.5 ± 1.4	89.9 ± 2.0	91.8 ± 1.5
Linear Ramp	89.4 ± 1.4	88.7 ± 2.5	87.4 ± 2.1	90.4 ± 2.2
Linear Ramp†	85.3 ± 1.9	76.7 ± 4.2	84.6 ± 2.4	87.2 ± 3.9
Recursive TS	90.1 ± 2.2	89.0 ± 2.8	88.3 ± 4.8	90.1 ± 3.5
TQA*	89.4 ± 1.9	88.8 ± 1.8	87.8 ± 2.7	91.0 ± 2.5
TQA	84.8 ± 3.8	81.8 ± 4.3	79.9 ± 6.0	84.0 ± 5.6

Table 11: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for MAXCUT for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	RR
MPS (Aer)								
Fixed Angles*	-	88.8 ± 1.3	82.8 ± 6.9	88.6 ± 8.8	-	10 [39]	21 [100]	13 [40-
Fixed Angles†	-	87.9 ± 1.5	77.1 ± 8.3	85.4 ± 11.0	-	10 [39]	21 [100]	13 [40-
Fourier*	75.6 ± 4.5	80.9 ± 2.7	74.1 ± 8.5	76.6 ± 5.5	20 [40–50]	20 [39–144]	56 [40–100]	17 [40-
Interp.*	87.0 ± 2.6	85.7 ± 3.4	86.2 ± 6.2	77.7 ± 6.7	20 [40–50]	20 [39–144]	56 [40–100]	17 [40-
Linear Ramp*	85.8 ± 4.1	81.4 ± 1.0	91.1 ± 1.2	75.0 ± 6.0	10 [50]	10 [144]	10 [100]	7 [10
Linear Ramp	81.7 ± 4.7	79.9 ± 1.7	88.8 ± 1.8	74.5 ± 5.8	10 [50]	10 [144]	10 [100]	7 [10
Linear Ramp†	75.4 ± 1.5	73.4 ± 1.5	82.5 ± 1.1	66.5 ± 3.4	10 [50]	10 [144]	10 [100]	7 [10
Recursive TS	84.7 ± 4.2	81.0 ± 4.1	87.7 ± 3.7	78.8 ± 4.9	20 [40–50]	20 [39–144]	50 [40–100]	11 [40-
TQA*	77.8 ± 4.9	82.5 ± 2.7	78.7 ± 9.6	79.2 ± 6.3	20 [40–50]	20 [39–144]	56 [40–100]	17 [40-
TQA	69.6 ± 1.8	79.8 ± 2.1	74.4 ± 9.2	75.4 ± 5.1	20 [40–50]	20 [39–144]	56 [40–100]	17 [40-
MPS (Quimb)								
Fixed Angles*	-	52.8 ± 1.1	90.5 ± 1.4	89.5 ± 5.9	-	10 [144]	30 [100]	13 [40-
Fixed Angles†	-	51.0 ± 0.9	90.4 ± 1.5	84.8 ± 10.7	-	10 [144]	30 [100]	13 [40-
Fourier*	77.4 ± 6.4	76.0 ± 8.2	83.0 ± 4.4	83.6 ± 9.6	20 [40–50]	31 [39–144]	130 [40–100]	28 [40-
Interp.*	92.1 ± 1.1	79.5 ± 8.6	89.9 ± 1.1	87.3 ± 8.4	20 [40–50]	31 [39–144]	130 [40–100]	28 [40-
Linear Ramp*	92.4 ± 1.9	68.8 ± 2.0	90.2 ± 1.0	77.8 ± 7.4	10 [50]	10 [144]	10 [100]	7 [10
Linear Ramp	90.7 ± 1.5	69.4 ± 2.0	87.7 ± 1.6	76.7 ± 5.8	10 [50]	10 [144]	10 [100]	7 [10
Linear Ramp†	83.6 ± 2.6	64.4 ± 1.2	83.4 ± 1.4	66.0 ± 4.1	10 [50]	10 [144]	10 [100]	7 [10
Recursive TS	92.6 ± 1.6	68.8 ± 12.3	88.7 ± 1.9	78.5 ± 7.4	20 [40–50]	31 [39–144]	10 [100]	18 [40-
TQA*	79.0 ± 7.0	70.7 ± 16.6	88.5 ± 1.9	84.2 ± 7.9	20 [40–50]	20 [39–144]	130 [40–100]	27 [40-
TQA	70.5 ± 3.4	68.1 ± 14.3	84.2 ± 2.4	77.4 ± 5.5	20 [40–50]	20 [39–144]	130 [40–100]	27 [40-
PP								
Fixed Angles*	-	89.3 ± 1.1	86.2 ± 1.7	89.6 ± 1.5	-	30 [39–144]	21 [100]	13 [40-
Fixed Angles†	-	88.2 ± 1.5	84.2 ± 1.4	89.4 ± 1.6	-	30 [39–144]	21 [100]	13 [40-
Fourier*	88.0 ± 2.1	86.4 ± 2.7	82.4 ± 2.9	87.2 ± 1.8	20 [40–50]	31 [39–144]	130 [40–100]	28 [40-
Interp.*	90.7 ± 1.9	89.3 ± 1.1	85.3 ± 1.8	89.1 ± 1.5	20 [40–50]	31 [39–144]	130 [40–100]	28 [40-
Linear Ramp*	92.9 ± 1.1	88.9 ± 0.8	86.6 ± 2.0	88.6 ± 0.7	10 [50]	20 [105–144]	20 [40–100]	8 [10
Linear Ramp	91.4 ± 1.0	87.5 ± 0.7	85.2 ± 2.3	87.9 ± 0.7	10 [50]	20 [105–144]	20 [40–100]	8 [10
Linear Ramp†	91.2 ± 1.0	77.1 ± 1.3	82.2 ± 1.0	84.5 ± 3.5	10 [50]	20 [105–144]	20 [40–100]	8 [10
Recursive TS	88.4 ± 1.2	85.4 ± 1.2	84.3 ± 1.8	85.4 ± 1.7	20 [40–50]	31 [39–144]	111 [40–100]	18 [40-
TQA*	89.2 ± 1.9	86.8 ± 1.2	83.5 ± 1.9	87.8 ± 1.4	20 [40–50]	20 [39–144]	130 [40–100]	27 [40-
TQA	83.7 ± 1.4	81.9 ± 1.3	75.7 ± 4.8	85.1 ± 1.8	20 [40–50]	20 [39–144]	130 [40–100]	27 [40-
SV								
Fixed Angles*	91.4 ± 1.6	-	91.7 ± 2.4	93.4 ± 2.2	61 [10–20]	-	106 [10–20]	155 [10
Fixed Angles†	89.0 ± 2.1	-	90.9 ± 2.4	92.8 ± 2.2	61 [10–20]	-	106 [10–20]	158 [10
Fourier*	78.4 ± 10.6	81.3 ± 6.1	80.7 ± 7.8	90.3 ± 4.4	92 [10–20]	20 [12–21]	784 [10–20]	567 [10
Interp.*	91.2 ± 1.6	91.3 ± 1.4	90.3 ± 3.8	90.9 ± 4.9	92 [10–20]	20 [12–21]	827 [10–22]	382 [10
Linear Ramp*	90.9 ± 1.2	90.5 ± 1.4	89.9 ± 2.0	91.8 ± 1.5	10 [20]	10 [21]	10 [20]	7 [2
Linear Ramp	89.4 ± 1.4	88.7 ± 2.5	87.4 ± 2.1	90.4 ± 2.2	10 [20]	10 [21]	10 [20]	7 [2
Linear Ramp†	85.3 ± 1.9	76.7 ± 4.2	84.6 ± 2.4	87.2 ± 3.9	10 [20]	10 [21]	10 [20]	7 [2
Recursive TS	90.1 ± 2.2	89.0 ± 2.8	88.3 ± 4.8	90.1 ± 3.5	92 [10–20]	20 [12–21]	504 [10–22]	381 [10
TQA*	89.4 ± 1.9	88.8 ± 1.8	87.8 ± 2.7	91.0 ± 2.5	91 [10–20]	20 [12–21]	821 [10–22]	142 [10
TQA	84.8 ± 3.8	81.8 ± 4.3	79.9 ± 6.0	84.0 ± 5.6	92 [10–20]	20 [12–21]	821 [10–22]	355 [10

Table 12: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

5 Tables for depth $P = 5$

	ER	HH	LB	RR
		MPS (Aer)		
Fixed Angles*	-	10 [144]	2 [100]	2 [100]
Fixed Angles†	-	10 [144]	2 [100]	2 [100]
Fourier*	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
Interp.*	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	19 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	19 [100]	7 [100]
Recursive TS	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
TQA*	10 [50]	10 [144]	10 [100]	7 [100]
TQA	10 [50]	10 [144]	10 [100]	7 [100]
		MPS (Quimb)		
Fixed Angles*	-	-	2 [100]	12 [40–100]
Fixed Angles†	-	-	2 [100]	12 [40–100]
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	28 [40–100]
Interp.*	20 [40–50]	31 [39–144]	130 [40–100]	28 [40–100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS	20 [40–50]	31 [39–144]	10 [100]	18 [40–100]
TQA*	10 [50]	20 [105–144]	20 [40–100]	17 [40–100]
TQA	10 [50]	20 [105–144]	20 [40–100]	17 [40–100]
		PP		
Fixed Angles*	-	10 [144]	2 [100]	12 [40–100]
Fixed Angles†	-	10 [144]	2 [100]	12 [40–100]
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	28 [40–100]
Interp.*	20 [40–50]	31 [39–144]	130 [40–100]	28 [40–100]
Linear Ramp*	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp	10 [50]	20 [105–144]	29 [40–100]	8 [100]
Linear Ramp†	10 [50]	20 [105–144]	29 [40–100]	8 [100]
Recursive TS	20 [40–50]	31 [39–144]	111 [40–100]	18 [40–100]
TQA*	10 [50]	20 [105–144]	20 [40–100]	17 [40–100]
TQA	10 [50]	20 [105–144]	20 [40–100]	17 [40–100]
		SV		
Fixed Angles*	46 [10–20]	-	60 [10–20]	115 [10–20]
Fixed Angles†	46 [10–20]	-	61 [10–20]	121 [10–20]
Fourier*	92 [10–20]	20 [12–21]	784 [10–20]	382 [10–20]
Interp.*	92 [10–20]	20 [12–21]	781 [10–20]	382 [10–20]
Linear Ramp*	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp†	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS	92 [10–20]	20 [12–21]	416 [10–20]	381 [10–20]
TQA*	90 [10–20]	20 [12–21]	672 [10–20]	142 [10–20]
TQA	92 [10–20]	20 [12–21]	677 [10–20]	359 [10–20]

Table 13: **Number of Instances (num. nodes in brackets) for** $P = 5$**.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angles⋆	-	84.3 ± 3.3	94.5 ± 0.7	79.1 ± 2.5
Fixed Angles†	-	83.3 ± 3.4	94.4 ± 0.6	68.1 ± 0.6
Fourier⋆	79.9 ± 6.1	82.4 ± 3.3	78.0 ± 10.4	76.8 ± 6.3
Interp.⋆	90.5 ± 2.0	87.2 ± 3.8	89.4 ± 5.3	80.8 ± 9.1
Linear Ramp⋆	88.8 ± 2.7	82.0 ± 2.0	93.4 ± 0.8	82.3 ± 5.4
Linear Ramp	85.1 ± 4.6	85.7 ± 1.0	78.4 ± 14.1	79.7 ± 5.5
Linear Ramp†	76.3 ± 2.5	77.6 ± 0.9	74.0 ± 13.8	66.0 ± 3.5
Recursive TS	87.8 ± 5.0	81.8 ± 4.3	90.5 ± 2.8	84.3 ± 6.2
TQA⋆	75.4 ± 2.6	81.5 ± 1.3	91.0 ± 1.2	80.1 ± 8.9
TQA	67.9 ± 7.7	79.2 ± 1.2	89.2 ± 1.8	74.9 ± 7.2
MPS (Quimb)				
Fixed Angles⋆	-	-	94.4 ± 0.7	93.2 ± 4.2
Fixed Angles†	-	-	94.3 ± 0.7	90.0 ± 8.9
Fourier⋆	82.0 ± 7.5	77.1 ± 8.9	85.8 ± 4.6	86.4 ± 10.4
Interp.⋆	94.5 ± 1.2	81.5 ± 8.7	92.4 ± 1.1	90.4 ± 7.1
Linear Ramp⋆	94.2 ± 1.5	70.2 ± 1.3	92.4 ± 1.0	86.3 ± 4.9
Linear Ramp	93.1 ± 1.3	69.3 ± 1.6	90.0 ± 1.5	84.8 ± 4.0
Linear Ramp†	87.4 ± 2.3	65.9 ± 1.0	86.2 ± 2.5	65.8 ± 3.8
Recursive TS	94.6 ± 1.3	69.9 ± 12.4	91.0 ± 1.4	83.1 ± 7.8
TQA⋆	77.8 ± 6.5	75.2 ± 4.8	90.9 ± 1.2	85.4 ± 8.1
TQA	66.5 ± 5.3	73.4 ± 4.6	87.8 ± 1.5	79.0 ± 7.8
PP				
Fixed Angles⋆	-	90.5 ± 0.8	89.1 ± 0.1	91.4 ± 1.6
Fixed Angles†	-	88.7 ± 1.8	86.3 ± 0.9	91.1 ± 1.7
Fourier⋆	89.6 ± 1.9	88.4 ± 2.7	83.5 ± 2.5	88.6 ± 1.6
Interp.⋆	91.9 ± 2.0	91.1 ± 1.0	86.6 ± 1.8	90.3 ± 1.8
Linear Ramp⋆	94.0 ± 1.0	90.2 ± 0.6	88.1 ± 2.1	89.6 ± 0.8
Linear Ramp	92.5 ± 1.0	89.0 ± 0.8	86.8 ± 2.1	88.9 ± 0.8
Linear Ramp†	92.4 ± 1.0	78.9 ± 1.3	83.6 ± 1.1	85.5 ± 3.1
Recursive TS	89.1 ± 1.3	86.8 ± 1.4	85.2 ± 1.9	86.7 ± 1.6
TQA⋆	93.3 ± 1.4	88.2 ± 0.8	84.1 ± 2.0	89.2 ± 1.4
TQA	83.6 ± 0.9	83.8 ± 1.1	72.8 ± 5.9	85.6 ± 1.5
SV				
Fixed Angles⋆	93.1 ± 1.4	-	94.8 ± 2.1	95.3 ± 2.0
Fixed Angles†	90.1 ± 2.2	-	93.7 ± 2.1	94.9 ± 2.0
Fourier⋆	80.5 ± 10.2	79.6 ± 6.2	81.2 ± 8.2	84.9 ± 10.6
Interp.⋆	92.9 ± 1.4	92.2 ± 1.0	92.4 ± 3.9	92.3 ± 4.5
Linear Ramp⋆	92.5 ± 1.1	92.2 ± 1.0	92.0 ± 1.9	93.4 ± 1.3
Linear Ramp	90.7 ± 1.3	90.8 ± 0.7	89.0 ± 1.9	91.5 ± 2.1
Linear Ramp†	86.5 ± 1.8	78.4 ± 4.1	86.1 ± 2.3	88.3 ± 3.7
Recursive TS	91.7 ± 2.0	90.5 ± 2.5	90.0 ± 5.1	91.2 ± 3.5
TQA⋆	91.6 ± 1.5	90.2 ± 1.8	90.6 ± 3.3	93.1 ± 2.2
TQA	86.9 ± 3.6	83.6 ± 4.3	81.9 ± 6.3	86.6 ± 5.3

Table 14: **Avg. Approx. Ratio ± standard deviation in percentage for MAXCUT for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	R
MPS (Aer)								
Fixed Angles*	-	84.3 ± 3.3	94.5 ± 0.7	79.1 ± 2.5	-	10 [144]	2 [100]	2 [1
Fixed Angles†	-	83.3 ± 3.4	94.4 ± 0.6	68.1 ± 0.6	-	10 [144]	2 [100]	2 [1
Fourier*	79.9 ± 6.1	82.4 ± 3.3	78.0 ± 10.4	76.8 ± 6.3	20 [40–50]	20 [39–144]	56 [40–100]	17 [40
Interp.*	90.5 ± 2.0	87.2 ± 3.8	89.4 ± 5.3	80.8 ± 9.1	20 [40–50]	20 [39–144]	56 [40–100]	17 [40
Linear Ramp*	88.8 ± 2.7	82.0 ± 2.0	93.4 ± 0.8	82.3 ± 5.4	10 [50]	10 [144]	10 [100]	7 [1
Linear Ramp	85.1 ± 4.6	85.7 ± 1.0	78.4 ± 14.1	79.7 ± 5.5	10 [50]	10 [144]	19 [100]	7 [1
Linear Ramp†	76.3 ± 2.5	77.6 ± 0.9	74.0 ± 13.8	66.0 ± 3.5	10 [50]	10 [144]	19 [100]	7 [1
Recursive TS	87.8 ± 5.0	81.8 ± 4.3	90.5 ± 2.8	84.3 ± 6.2	20 [40–50]	20 [39–144]	50 [40–100]	11 [40
TQA*	75.4 ± 2.6	81.5 ± 1.3	91.0 ± 1.2	80.1 ± 8.9	10 [50]	10 [144]	10 [100]	7 [1
TQA	67.9 ± 7.7	79.2 ± 1.2	89.2 ± 1.8	74.9 ± 7.2	10 [50]	10 [144]	10 [100]	7 [1
MPS (Quimb)								
Fixed Angles*	-	-	94.4 ± 0.7	93.2 ± 4.2	-	-	2 [100]	12 [40
Fixed Angles†	-	-	94.3 ± 0.7	90.0 ± 8.9	-	-	2 [100]	12 [40
Fourier*	82.0 ± 7.5	77.1 ± 8.9	85.8 ± 4.6	86.4 ± 10.4	20 [40–50]	31 [39–144]	130 [40–100]	28 [40
Interp.*	94.5 ± 1.2	81.5 ± 8.7	92.4 ± 1.1	90.4 ± 7.1	20 [40–50]	31 [39–144]	130 [40–100]	28 [40
Linear Ramp*	94.2 ± 1.5	70.2 ± 1.3	92.4 ± 1.0	86.3 ± 4.9	10 [50]	10 [144]	10 [100]	7 [1
Linear Ramp	93.1 ± 1.3	69.3 ± 1.6	90.0 ± 1.5	84.8 ± 4.0	10 [50]	10 [144]	10 [100]	7 [1
Linear Ramp†	87.4 ± 2.3	65.9 ± 1.0	86.2 ± 2.5	65.8 ± 3.8	10 [50]	10 [144]	10 [100]	7 [1
Recursive TS	94.6 ± 1.3	69.9 ± 12.4	91.0 ± 1.4	83.1 ± 7.8	20 [40–50]	31 [39–144]	10 [100]	18 [40
TQA*	77.8 ± 6.5	75.2 ± 4.8	90.9 ± 1.2	85.4 ± 8.1	10 [50]	20 [105–144]	20 [40–100]	17 [40
TQA	66.5 ± 5.3	73.4 ± 4.6	87.8 ± 1.5	79.0 ± 7.8	10 [50]	20 [105–144]	20 [40–100]	17 [40
PP								
Fixed Angles*	-	90.5 ± 0.8	89.1 ± 0.1	91.4 ± 1.6	-	10 [144]	2 [100]	12 [40
Fixed Angles†	-	88.7 ± 1.8	86.3 ± 0.9	91.1 ± 1.7	-	10 [144]	2 [100]	12 [40
Fourier*	89.6 ± 1.9	88.4 ± 2.7	83.5 ± 2.5	88.6 ± 1.6	20 [40–50]	31 [39–144]	130 [40–100]	28 [40
Interp.*	91.9 ± 2.0	91.1 ± 1.0	86.6 ± 1.8	90.3 ± 1.8	20 [40–50]	31 [39–144]	130 [40–100]	28 [40
Linear Ramp*	94.0 ± 1.0	90.2 ± 0.6	88.1 ± 2.1	89.6 ± 0.8	10 [50]	20 [105–144]	20 [40–100]	8 [1
Linear Ramp	92.5 ± 1.0	89.0 ± 0.8	86.8 ± 2.1	88.9 ± 0.8	10 [50]	20 [105–144]	29 [40–100]	8 [1
Linear Ramp†	92.4 ± 1.0	78.9 ± 1.3	83.6 ± 1.1	85.5 ± 3.1	10 [50]	20 [105–144]	29 [40–100]	8 [1
Recursive TS	89.1 ± 1.3	86.8 ± 1.4	85.2 ± 1.9	86.7 ± 1.6	20 [40–50]	31 [39–144]	111 [40–100]	18 [40
TQA*	93.3 ± 1.4	88.2 ± 0.8	84.1 ± 2.0	89.2 ± 1.4	10 [50]	20 [105–144]	20 [40–100]	17 [40
TQA	83.6 ± 0.9	83.8 ± 1.1	72.8 ± 5.9	85.6 ± 1.5	10 [50]	20 [105–144]	20 [40–100]	17 [40
SV								
Fixed Angles*	93.1 ± 1.4	-	94.8 ± 2.1	95.3 ± 2.0	46 [10–20]	-	60 [10–20]	115 [1
Fixed Angles†	90.1 ± 2.2	-	93.7 ± 2.1	94.9 ± 2.0	46 [10–20]	-	61 [10–20]	121 [1
Fourier*	80.5 ± 10.2	79.6 ± 6.2	81.2 ± 8.2	84.9 ± 10.6	92 [10–20]	20 [12–21]	784 [10–20]	382 [1
Interp.*	92.9 ± 1.4	92.2 ± 1.0	92.4 ± 3.9	92.3 ± 4.5	92 [10–20]	20 [12–21]	781 [10–20]	382 [1
Linear Ramp*	92.5 ± 1.1	92.2 ± 1.0	92.0 ± 1.9	93.4 ± 1.3	10 [20]	10 [21]	10 [20]	7 [
Linear Ramp	90.7 ± 1.3	90.8 ± 0.7	89.0 ± 1.9	91.5 ± 2.1	10 [20]	10 [21]	10 [20]	7 [
Linear Ramp†	86.5 ± 1.8	78.4 ± 4.1	86.1 ± 2.3	88.3 ± 3.7	10 [20]	10 [21]	10 [20]	7 [
Recursive TS	91.7 ± 2.0	90.5 ± 2.5	90.0 ± 5.1	91.2 ± 3.5	92 [10–20]	20 [12–21]	416 [10–20]	381 [1
TQA*	91.6 ± 1.5	90.2 ± 1.8	90.6 ± 3.3	93.1 ± 2.2	90 [10–20]	20 [12–21]	672 [10–20]	142 [1
TQA	86.9 ± 3.6	83.6 ± 4.3	81.9 ± 6.3	86.6 ± 5.3	92 [10–20]	20 [12–21]	677 [10–20]	359 [1

Table 15: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

6 Tables for depth $P = 6$

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angles*	-	10 [39]	20 [100]	11 [40–100]
Fixed Angles†	-	10 [39]	20 [100]	11 [40–100]
Fourier*	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
Interp.*	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	19 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	19 [100]	7 [100]
Recursive TS	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
TQA*	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
TQA	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
MPS (Quimb)				
Fixed Angles*	-	10 [144]	10 [100]	11 [40–100]
Fixed Angles†	-	10 [144]	10 [100]	11 [40–100]
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	28 [40–100]
Interp.*	20 [40–50]	31 [39–144]	130 [40–100]	28 [40–100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS	20 [40–50]	31 [39–144]	10 [100]	18 [40–100]
TQA*	20 [40–50]	20 [39–144]	130 [40–100]	27 [40–100]
TQA	20 [40–50]	20 [39–144]	130 [40–100]	27 [40–100]
PP				
Fixed Angles*	-	30 [39–144]	10 [100]	11 [40–100]
Fixed Angles†	-	30 [39–144]	10 [100]	11 [40–100]
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	28 [40–100]
Interp.*	20 [40–50]	31 [39–144]	130 [40–100]	28 [40–100]
Linear Ramp*	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp	10 [50]	20 [105–144]	29 [40–100]	8 [100]
Linear Ramp†	10 [50]	20 [105–144]	29 [40–100]	8 [100]
Recursive TS	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–100]
TQA*	20 [40–50]	20 [39–144]	130 [40–100]	27 [40–100]
TQA	20 [40–50]	20 [39–144]	130 [40–100]	27 [40–100]
SV				
Fixed Angles*	17 [11–20]	-	20 [10–20]	59 [10–20]
Fixed Angles†	17 [11–20]	-	20 [10–20]	59 [10–20]
Fourier*	92 [10–20]	20 [12–21]	781 [10–20]	382 [10–20]
Interp.*	92 [10–20]	20 [12–21]	781 [10–20]	382 [10–20]
Linear Ramp*	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp†	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS	92 [10–20]	20 [12–21]	413 [10–20]	381 [10–20]
TQA*	90 [10–20]	20 [12–21]	672 [10–20]	142 [10–20]
TQA	92 [10–20]	20 [12–21]	677 [10–20]	360 [10–20]

Table 16: **Number of Instances (num. nodes in brackets) for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
		MPS (Aer)		
Fixed Angles*	-	91.6 ± 0.9	90.1 ± 2.9	95.6 ± 5.3
Fixed Angles†	-	89.9 ± 1.7	86.0 ± 5.5	94.3 ± 7.6
Fourier*	78.1 ± 5.7	83.1 ± 3.0	79.9 ± 9.2	81.4 ± 7.6
Interp.*	92.2 ± 1.8	88.3 ± 4.4	91.9 ± 3.8	84.7 ± 7.0
Linear Ramp*	92.0 ± 1.8	81.4 ± 1.7	94.6 ± 0.9	86.4 ± 2.2
Linear Ramp	89.2 ± 1.8	81.4 ± 2.7	79.7 ± 14.0	84.5 ± 3.1
Linear Ramp†	78.8 ± 2.8	74.8 ± 2.1	75.4 ± 14.4	65.8 ± 3.5
Recursive TS	89.8 ± 5.4	83.1 ± 4.7	92.0 ± 2.5	88.3 ± 6.8
TQA*	80.9 ± 3.8	84.6 ± 3.1	84.5 ± 8.8	87.8 ± 8.6
TQA	74.3 ± 4.5	82.9 ± 2.5	80.2 ± 8.6	82.2 ± 7.4
		MPS (Quimb)		
Fixed Angles*	-	53.9 ± 2.0	95.1 ± 1.1	96.1 ± 2.4
Fixed Angles†	-	50.9 ± 1.1	95.1 ± 1.0	94.7 ± 4.5
Fourier*	78.9 ± 7.6	76.6 ± 10.0	87.9 ± 3.5	88.9 ± 9.6
Interp.*	95.8 ± 1.1	81.9 ± 9.7	93.9 ± 1.0	92.0 ± 6.2
Linear Ramp*	95.2 ± 1.6	69.2 ± 1.6	93.9 ± 1.3	86.4 ± 7.3
Linear Ramp	94.1 ± 1.3	67.9 ± 1.8	91.4 ± 1.3	84.3 ± 7.6
Linear Ramp†	90.0 ± 2.0	65.3 ± 1.2	88.0 ± 2.9	65.4 ± 4.2
Recursive TS	95.8 ± 0.8	69.7 ± 13.3	92.2 ± 1.4	85.2 ± 8.1
TQA*	85.5 ± 6.7	71.7 ± 17.6	91.1 ± 1.4	90.6 ± 7.1
TQA	71.7 ± 6.0	70.0 ± 16.1	88.0 ± 2.1	85.9 ± 8.9
		PP		
Fixed Angles*	-	92.0 ± 0.7	89.1 ± 1.8	92.3 ± 1.6
Fixed Angles†	-	90.2 ± 1.5	85.5 ± 1.6	91.9 ± 1.7
Fourier*	90.7 ± 1.9	89.3 ± 2.8	84.8 ± 2.2	89.5 ± 1.8
Interp.*	92.7 ± 2.1	92.2 ± 0.9	87.6 ± 1.8	91.1 ± 2.0
Linear Ramp*	94.7 ± 1.0	91.3 ± 0.7	89.2 ± 2.2	90.4 ± 0.7
Linear Ramp	93.2 ± 1.0	90.0 ± 0.7	87.8 ± 2.0	89.5 ± 0.8
Linear Ramp†	93.1 ± 1.0	80.5 ± 1.2	84.7 ± 1.1	86.4 ± 2.9
Recursive TS	89.5 ± 1.2	87.6 ± 1.6	85.8 ± 2.1	87.3 ± 1.5
TQA*	91.3 ± 2.4	89.6 ± 0.9	85.2 ± 2.2	90.3 ± 1.6
TQA	84.6 ± 1.9	85.6 ± 1.2	76.5 ± 5.9	87.5 ± 2.1
		SV		
Fixed Angles*	94.4 ± 1.5	-	96.5 ± 1.4	96.2 ± 1.4
Fixed Angles†	90.3 ± 2.5	-	95.6 ± 1.3	95.9 ± 1.3
Fourier*	80.3 ± 10.6	78.5 ± 9.3	80.9 ± 9.2	84.6 ± 11.5
Interp.*	94.1 ± 1.3	93.2 ± 1.2	93.7 ± 3.8	92.9 ± 4.9
Linear Ramp*	93.5 ± 1.0	93.0 ± 0.9	93.2 ± 1.9	94.5 ± 1.0
Linear Ramp	91.8 ± 1.3	91.4 ± 1.3	90.1 ± 1.7	92.4 ± 1.9
Linear Ramp†	87.6 ± 1.7	80.2 ± 4.1	87.3 ± 2.2	89.2 ± 3.5
Recursive TS	92.9 ± 1.9	91.2 ± 2.4	91.3 ± 5.0	92.2 ± 3.4
TQA*	92.9 ± 1.4	90.8 ± 1.9	91.9 ± 3.0	94.2 ± 2.1
TQA	88.3 ± 3.3	85.4 ± 3.9	82.5 ± 7.3	88.4 ± 5.0

Table 17: **Avg. Approx. Ratio ± standard deviation in percentage for MAXCUT for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	R
MPS (Aer)								
Fixed Angles*	-	91.6 ± 0.9	90.1 ± 2.9	95.6 ± 5.3	-	10 [39]	20 [100]	11 [40
Fixed Angles†	-	89.9 ± 1.7	86.0 ± 5.5	94.3 ± 7.6	-	10 [39]	20 [100]	11 [40
Fourier*	78.1 ± 5.7	83.1 ± 3.0	79.9 ± 9.2	81.4 ± 7.6	20 [40–50]	20 [39–144]	56 [40–100]	17 [40
Interp.*	92.2 ± 1.8	88.3 ± 4.4	91.9 ± 3.8	84.7 ± 7.0	20 [40–50]	20 [39–144]	56 [40–100]	17 [40
Linear Ramp*	92.0 ± 1.8	81.4 ± 1.7	94.6 ± 0.9	86.4 ± 2.2	10 [50]	10 [144]	10 [100]	7 [1
Linear Ramp	89.2 ± 1.8	81.4 ± 2.7	79.7 ± 14.0	84.5 ± 3.1	10 [50]	10 [144]	19 [100]	7 [1
Linear Ramp†	78.8 ± 2.8	74.8 ± 2.1	75.4 ± 14.4	65.8 ± 3.5	10 [50]	10 [144]	19 [100]	7 [1
Recursive TS	89.8 ± 5.4	83.1 ± 4.7	92.0 ± 2.5	88.3 ± 6.8	20 [40–50]	20 [39–144]	50 [40–100]	11 [40
TQA*	80.9 ± 3.8	84.6 ± 3.1	84.5 ± 8.8	87.8 ± 8.6	20 [40–50]	20 [39–144]	56 [40–100]	17 [40
TQA	74.3 ± 4.5	82.9 ± 2.5	80.2 ± 8.6	82.2 ± 7.4	20 [40–50]	20 [39–144]	56 [40–100]	17 [40
MPS (Quimb)								
Fixed Angles*	-	53.9 ± 2.0	95.1 ± 1.1	96.1 ± 2.4	-	10 [144]	10 [100]	11 [40
Fixed Angles†	-	50.9 ± 1.1	95.1 ± 1.0	94.7 ± 4.5	-	10 [144]	10 [100]	11 [40
Fourier*	78.9 ± 7.6	76.6 ± 10.0	87.9 ± 3.5	88.9 ± 9.6	20 [40–50]	31 [39–144]	130 [40–100]	28 [40
Interp.*	95.8 ± 1.1	81.9 ± 9.7	93.9 ± 1.0	92.0 ± 6.2	20 [40–50]	31 [39–144]	130 [40–100]	28 [40
Linear Ramp*	95.2 ± 1.6	69.2 ± 1.6	93.9 ± 1.3	86.4 ± 7.3	10 [50]	10 [144]	10 [100]	7 [1
Linear Ramp	94.1 ± 1.3	67.9 ± 1.8	91.4 ± 1.3	84.3 ± 7.6	10 [50]	10 [144]	10 [100]	7 [1
Linear Ramp†	90.0 ± 2.0	65.3 ± 1.2	88.0 ± 2.9	65.4 ± 4.2	10 [50]	10 [144]	10 [100]	7 [1
Recursive TS	95.8 ± 0.8	69.7 ± 13.3	92.2 ± 1.4	85.2 ± 8.1	20 [40–50]	31 [39–144]	10 [100]	18 [40
TQA*	85.5 ± 6.7	71.7 ± 17.6	91.1 ± 1.4	90.6 ± 7.1	20 [40–50]	20 [39–144]	130 [40–100]	27 [40
TQA	71.7 ± 6.0	70.0 ± 16.1	88.0 ± 2.1	85.9 ± 8.9	20 [40–50]	20 [39–144]	130 [40–100]	27 [40
PP								
Fixed Angles*	-	92.0 ± 0.7	89.1 ± 1.8	92.3 ± 1.6	-	30 [39–144]	10 [100]	11 [40
Fixed Angles†	-	90.2 ± 1.5	85.5 ± 1.6	91.9 ± 1.7	-	30 [39–144]	10 [100]	11 [40
Fourier*	90.7 ± 1.9	89.3 ± 2.8	84.8 ± 2.2	89.5 ± 1.8	20 [40–50]	31 [39–144]	130 [40–100]	28 [40
Interp.*	92.7 ± 2.1	92.2 ± 0.9	87.6 ± 1.8	91.1 ± 2.0	20 [40–50]	31 [39–144]	130 [40–100]	28 [40
Linear Ramp*	94.7 ± 1.0	91.3 ± 0.7	89.2 ± 2.2	90.4 ± 0.7	10 [50]	20 [105–144]	20 [40–100]	8 [1
Linear Ramp	93.2 ± 1.0	90.0 ± 0.7	87.8 ± 2.0	89.5 ± 0.8	10 [50]	20 [105–144]	29 [40–100]	8 [1
Linear Ramp†	93.1 ± 1.0	80.5 ± 1.2	84.7 ± 1.1	86.4 ± 2.9	10 [50]	20 [105–144]	29 [40–100]	8 [1
Recursive TS	89.5 ± 1.2	87.6 ± 1.6	85.8 ± 2.1	87.3 ± 1.5	20 [40–50]	30 [39–144]	111 [40–100]	18 [40
TQA*	91.3 ± 2.4	89.6 ± 0.9	85.2 ± 2.2	90.3 ± 1.6	20 [40–50]	20 [39–144]	130 [40–100]	27 [40
TQA	84.6 ± 1.9	85.6 ± 1.2	76.5 ± 5.9	87.5 ± 2.1	20 [40–50]	20 [39–144]	130 [40–100]	27 [40
SV								
Fixed Angles*	94.4 ± 1.5	-	96.5 ± 1.4	96.2 ± 1.4	17 [11–20]	-	20 [10–20]	59 [1
Fixed Angles†	90.3 ± 2.5	-	95.6 ± 1.3	95.9 ± 1.3	17 [11–20]	-	20 [10–20]	59 [1
Fourier*	80.3 ± 10.6	78.5 ± 9.3	80.9 ± 9.2	84.6 ± 11.5	92 [10–20]	20 [12–21]	781 [10–20]	382 [
Interp.*	94.1 ± 1.3	93.2 ± 1.2	93.7 ± 3.8	92.9 ± 4.9	92 [10–20]	20 [12–21]	781 [10–20]	382 [
Linear Ramp*	93.5 ± 1.0	93.0 ± 0.9	93.2 ± 1.9	94.5 ± 1.0	10 [20]	10 [21]	10 [20]	7 [
Linear Ramp	91.8 ± 1.3	91.4 ± 1.3	90.1 ± 1.7	92.4 ± 1.9	10 [20]	10 [21]	10 [20]	7 [
Linear Ramp†	87.6 ± 1.7	80.2 ± 4.1	87.3 ± 2.2	89.2 ± 3.5	10 [20]	10 [21]	10 [20]	7 [
Recursive TS	92.9 ± 1.9	91.2 ± 2.4	91.3 ± 5.0	92.2 ± 3.4	92 [10–20]	20 [12–21]	413 [10–20]	381 [
TQA*	92.9 ± 1.4	90.8 ± 1.9	91.9 ± 3.0	94.2 ± 2.1	90 [10–20]	20 [12–21]	672 [10–20]	142 [
TQA	88.3 ± 3.3	85.4 ± 3.9	82.5 ± 7.3	88.4 ± 5.0	92 [10–20]	20 [12–21]	677 [10–20]	360 [

Table 18: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

7 Tables for depth $P = 7$

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angles*	-	-	1 [100]	1 [100]
Fixed Angles†	-	-	1 [100]	1 [100]
Fourier*	20 [40–50]	20 [39–144]	50 [40–100]	17 [40–100]
Interp.*	20 [40–50]	20 [39–144]	50 [40–100]	17 [40–100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
TQA*	10 [50]	10 [144]	10 [100]	7 [100]
TQA	10 [50]	10 [144]	10 [100]	7 [100]
MPS (Quimb)				
Fixed Angles*	-	10 [144]	1 [100]	1 [100]
Fixed Angles†	-	10 [144]	1 [100]	1 [100]
Fourier*	20 [40–50]	31 [39–144]	120 [40–100]	28 [40–100]
Interp.*	20 [40–50]	31 [39–144]	120 [40–100]	28 [40–100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS	20 [40–50]	31 [39–144]	10 [100]	18 [40–100]
TQA*	10 [50]	10 [144]	10 [100]	7 [100]
TQA	10 [50]	10 [144]	10 [100]	7 [100]
PP				
Fixed Angles*	-	10 [144]	1 [100]	1 [100]
Fixed Angles†	-	10 [144]	1 [100]	1 [100]
Fourier*	20 [40–50]	31 [39–144]	120 [40–100]	28 [40–100]
Interp.*	20 [40–50]	31 [39–144]	120 [40–100]	28 [40–100]
Linear Ramp*	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp†	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Recursive TS	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–100]
TQA*	10 [50]	10 [144]	10 [100]	7 [100]
TQA	10 [50]	10 [144]	10 [100]	7 [100]
SV				
Fixed Angles*	17 [11–20]	-	19 [10–20]	59 [10–20]
Fixed Angles†	17 [11–20]	-	19 [10–20]	60 [10–20]
Fourier*	92 [10–20]	20 [12–21]	780 [10–20]	382 [10–20]
Interp.*	92 [10–20]	20 [12–21]	780 [10–20]	382 [10–20]
Linear Ramp*	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp†	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS	92 [10–20]	20 [12–21]	412 [10–20]	379 [10–20]
TQA*	90 [10–20]	20 [12–21]	672 [10–20]	141 [10–20]
TQA	92 [10–20]	20 [12–21]	677 [10–20]	361 [10–20]

Table 19: **Number of Instances (num. nodes in brackets) for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
		MPS (Aer)		
Fixed Angles*	-	-	97.0	82.6
Fixed Angles†	-	-	96.8	71.9
Fourier*	79.4 ± 5.7	84.4 ± 3.2	80.5 ± 9.4	83.7 ± 8.2
Interp.*	93.7 ± 2.1	89.2 ± 4.7	93.5 ± 3.0	85.0 ± 8.3
Linear Ramp*	92.9 ± 2.0	83.5 ± 2.5	95.2 ± 0.8	87.1 ± 3.3
Linear Ramp	89.8 ± 2.9	82.5 ± 2.2	93.4 ± 1.0	86.0 ± 2.4
Linear Ramp†	81.6 ± 3.1	76.3 ± 2.0	90.0 ± 3.0	65.8 ± 3.7
Recursive TS	91.6 ± 5.5	83.9 ± 4.6	93.0 ± 2.4	90.0 ± 7.3
TQA*	81.7 ± 2.5	83.2 ± 1.9	92.5 ± 1.2	86.8 ± 4.5
TQA	78.1 ± 5.3	82.0 ± 1.5	90.7 ± 1.1	79.2 ± 7.8
		MPS (Quimb)		
Fixed Angles*	-	56.3 ± 2.6	97.0	91.9
Fixed Angles†	-	52.3 ± 2.6	96.7	84.8
Fourier*	81.1 ± 7.1	77.6 ± 10.1	88.1 ± 4.1	89.0 ± 8.8
Interp.*	96.4 ± 1.3	83.7 ± 9.4	94.9 ± 1.0	93.0 ± 6.0
Linear Ramp*	95.9 ± 1.4	71.7 ± 1.7	94.6 ± 0.8	90.2 ± 2.5
Linear Ramp	94.8 ± 1.3	69.2 ± 1.5	92.5 ± 1.3	88.6 ± 3.4
Linear Ramp†	91.9 ± 1.6	66.4 ± 1.3	89.4 ± 3.0	65.6 ± 4.4
Recursive TS	96.5 ± 0.8	70.6 ± 12.8	93.0 ± 1.1	87.9 ± 7.0
TQA*	86.8 ± 4.8	57.5 ± 1.0	92.4 ± 0.8	83.2 ± 8.5
TQA	71.9 ± 9.7	57.3 ± 1.0	89.6 ± 1.0	76.8 ± 11.9
		PP		
Fixed Angles*	-	92.5 ± 0.6	90.6	90.7
Fixed Angles†	-	90.2 ± 1.6	86.4	89.8
Fourier*	90.0 ± 2.8	89.9 ± 2.7	85.2 ± 2.1	89.9 ± 1.7
Interp.*	93.3 ± 2.1	93.1 ± 0.9	88.3 ± 1.8	91.2 ± 2.1
Linear Ramp*	95.1 ± 1.0	92.1 ± 0.6	89.8 ± 2.3	90.8 ± 0.8
Linear Ramp	93.8 ± 1.0	90.9 ± 0.6	88.5 ± 2.2	90.0 ± 0.8
Linear Ramp†	93.5 ± 1.0	82.0 ± 1.2	85.8 ± 1.1	87.1 ± 2.6
Recursive TS	89.9 ± 1.2	88.3 ± 1.6	86.3 ± 2.1	87.8 ± 1.5
TQA*	94.5 ± 1.5	90.2 ± 0.6	85.6 ± 2.3	89.6 ± 1.1
TQA	83.8 ± 0.9	87.0 ± 0.8	77.2 ± 6.9	86.3 ± 2.3
		SV		
Fixed Angles*	95.4 ± 1.5	-	97.4 ± 1.2	97.2 ± 1.2
Fixed Angles†	90.7 ± 3.0	-	96.3 ± 1.0	96.8 ± 1.2
Fourier*	80.4 ± 10.8	77.9 ± 10.7	80.8 ± 9.7	84.1 ± 12.4
Interp.*	95.1 ± 1.2	93.9 ± 1.2	94.5 ± 4.0	93.1 ± 5.4
Linear Ramp*	94.3 ± 1.0	93.6 ± 0.8	94.0 ± 1.6	95.2 ± 0.9
Linear Ramp	92.5 ± 1.2	92.0 ± 1.3	91.1 ± 1.6	93.1 ± 1.8
Linear Ramp†	88.6 ± 1.6	81.7 ± 4.0	88.4 ± 2.1	90.0 ± 3.3
Recursive TS	93.8 ± 1.8	91.8 ± 2.3	92.3 ± 4.9	92.9 ± 3.3
TQA*	93.8 ± 1.3	91.4 ± 1.9	92.5 ± 3.2	95.0 ± 1.8
TQA	89.4 ± 3.3	86.8 ± 3.7	83.1 ± 7.8	89.3 ± 4.8

Table 20: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for MAXCUT for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	RR
MPS (Aer)								
Fixed Angles*	-	-	97.0	82.6	-	-	1 [100]	1 [10
Fixed Angles†	-	-	96.8	71.9	-	-	1 [100]	1 [10
Fourier*	79.4 ± 5.7	84.4 ± 3.2	80.5 ± 9.4	83.7 ± 8.2	20 [40–50]	20 [39–144]	50 [40–100]	17 [40–
Interp.*	93.7 ± 2.1	89.2 ± 4.7	93.5 ± 3.0	85.0 ± 8.3	20 [40–50]	20 [39–144]	50 [40–100]	17 [40–
Linear Ramp*	92.9 ± 2.0	83.5 ± 2.5	95.2 ± 0.8	87.1 ± 3.3	10 [50]	10 [144]	10 [100]	7 [10
Linear Ramp	89.8 ± 2.9	82.5 ± 2.2	93.4 ± 1.0	86.0 ± 2.4	10 [50]	10 [144]	10 [100]	7 [10
Linear Ramp†	81.6 ± 3.1	76.3 ± 2.0	90.0 ± 3.0	65.8 ± 3.7	10 [50]	10 [144]	10 [100]	7 [10
Recursive TS	91.6 ± 5.5	83.9 ± 4.6	93.0 ± 2.4	90.0 ± 7.3	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–
TQA*	81.7 ± 2.5	83.2 ± 1.9	92.5 ± 1.2	86.8 ± 4.5	10 [50]	10 [144]	10 [100]	7 [10
TQA	78.1 ± 5.3	82.0 ± 1.5	90.7 ± 1.1	79.2 ± 7.8	10 [50]	10 [144]	10 [100]	7 [10
MPS (Quimb)								
Fixed Angles*	-	56.3 ± 2.6	97.0	91.9	-	10 [144]	1 [100]	1 [10
Fixed Angles†	-	52.3 ± 2.6	96.7	84.8	-	10 [144]	1 [100]	1 [10
Fourier*	81.1 ± 7.1	77.6 ± 10.1	88.1 ± 4.1	89.0 ± 8.8	20 [40–50]	31 [39–144]	120 [40–100]	28 [40–
Interp.*	96.4 ± 1.3	83.7 ± 9.4	94.9 ± 1.0	93.0 ± 6.0	20 [40–50]	31 [39–144]	120 [40–100]	28 [40–
Linear Ramp*	95.9 ± 1.4	71.7 ± 1.7	94.6 ± 0.8	90.2 ± 2.5	10 [50]	10 [144]	10 [100]	7 [10
Linear Ramp	94.8 ± 1.3	69.2 ± 1.5	92.5 ± 1.3	88.6 ± 3.4	10 [50]	10 [144]	10 [100]	7 [10
Linear Ramp†	91.9 ± 1.6	66.4 ± 1.3	89.4 ± 3.0	65.6 ± 4.4	10 [50]	10 [144]	10 [100]	7 [10
Recursive TS	96.5 ± 0.8	70.6 ± 12.8	93.0 ± 1.1	87.9 ± 7.0	20 [40–50]	31 [39–144]	10 [100]	18 [40–
TQA*	86.8 ± 4.8	57.5 ± 1.0	92.4 ± 0.8	83.2 ± 8.5	10 [50]	10 [144]	10 [100]	7 [10
TQA	71.9 ± 9.7	57.3 ± 1.0	89.6 ± 1.0	76.8 ± 11.9	10 [50]	10 [144]	10 [100]	7 [10
PP								
Fixed Angles*	-	92.5 ± 0.6	90.6	90.7	-	10 [144]	1 [100]	1 [10
Fixed Angles†	-	90.2 ± 1.6	86.4	89.8	-	10 [144]	1 [100]	1 [10
Fourier*	90.0 ± 2.8	89.9 ± 2.7	85.2 ± 2.1	89.9 ± 1.7	20 [40–50]	31 [39–144]	120 [40–100]	28 [40–
Interp.*	93.3 ± 2.1	93.1 ± 0.9	88.3 ± 1.8	91.2 ± 2.1	20 [40–50]	31 [39–144]	120 [40–100]	28 [40–
Linear Ramp*	95.1 ± 1.0	92.1 ± 0.6	89.8 ± 2.3	90.8 ± 0.8	10 [50]	20 [105–144]	20 [40–100]	8 [10
Linear Ramp	93.8 ± 1.0	90.9 ± 0.6	88.5 ± 2.2	90.0 ± 0.8	10 [50]	20 [105–144]	20 [40–100]	8 [10
Linear Ramp†	93.5 ± 1.0	82.0 ± 1.2	85.8 ± 1.1	87.1 ± 2.6	10 [50]	20 [105–144]	20 [40–100]	8 [10
Recursive TS	89.9 ± 1.2	88.3 ± 1.6	86.3 ± 2.1	87.8 ± 1.5	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–
TQA*	94.5 ± 1.5	90.2 ± 0.6	85.6 ± 2.3	89.6 ± 1.1	10 [50]	10 [144]	10 [100]	7 [10
TQA	83.8 ± 0.9	87.0 ± 0.8	77.2 ± 6.9	86.3 ± 2.3	10 [50]	10 [144]	10 [100]	7 [10
SV								
Fixed Angles*	95.4 ± 1.5	-	97.4 ± 1.2	97.2 ± 1.2	17 [11–20]	-	19 [10–20]	59 [10
Fixed Angles†	90.7 ± 3.0	-	96.3 ± 1.0	96.8 ± 1.2	17 [11–20]	-	19 [10–20]	60 [10
Fourier*	80.4 ± 10.8	77.9 ± 10.7	80.8 ± 9.7	84.1 ± 12.4	92 [10–20]	20 [12–21]	780 [10–20]	382 [10
Interp.*	95.1 ± 1.2	93.9 ± 1.2	94.5 ± 4.0	93.1 ± 5.4	92 [10–20]	20 [12–21]	780 [10–20]	382 [10
Linear Ramp*	94.3 ± 1.0	93.6 ± 0.8	94.0 ± 1.6	95.2 ± 0.9	10 [20]	10 [21]	10 [20]	7 [2
Linear Ramp	92.5 ± 1.2	92.0 ± 1.3	91.1 ± 1.6	93.1 ± 1.8	10 [20]	10 [21]	10 [20]	7 [2
Linear Ramp†	88.6 ± 1.6	81.7 ± 4.0	88.4 ± 2.1	90.0 ± 3.3	10 [20]	10 [21]	10 [20]	7 [2
Recursive TS	93.8 ± 1.8	91.8 ± 2.3	92.3 ± 4.9	92.9 ± 3.3	92 [10–20]	20 [12–21]	412 [10–20]	379 [10
TQA*	93.8 ± 1.3	91.4 ± 1.9	92.5 ± 3.2	95.0 ± 1.8	90 [10–20]	20 [12–21]	672 [10–20]	141 [10
TQA	89.4 ± 3.3	86.8 ± 3.7	83.1 ± 7.8	89.3 ± 4.8	92 [10–20]	20 [12–21]	677 [10–20]	361 [10

Table 21: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

8 Tables for depth $P = 8$

	ER	HH	LB	RR
		MPS (Aer)		
Fixed Angles*	-	10 [39]	11 [100]	11 [40–100]
Fixed Angles†	-	10 [39]	11 [100]	11 [40–100]
Fourier*	20 [40–50]	20 [39–144]	50 [40–100]	17 [40–100]
Interp.*	20 [40–50]	20 [39–144]	50 [40–100]	17 [40–100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
TQA*	20 [40–50]	20 [39–144]	50 [40–100]	17 [40–100]
TQA	20 [40–50]	20 [39–144]	50 [40–100]	17 [40–100]
		MPS (Quimb)		
Fixed Angles*	-	10 [144]	10 [100]	11 [40–100]
Fixed Angles†	-	10 [144]	10 [100]	11 [40–100]
Fourier*	20 [40–50]	31 [39–144]	120 [40–100]	28 [40–100]
Interp.*	20 [40–50]	31 [39–144]	120 [40–100]	28 [40–100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS	20 [40–50]	31 [39–144]	10 [100]	18 [40–100]
TQA*	20 [40–50]	20 [39–144]	120 [40–100]	27 [40–100]
TQA	20 [40–50]	20 [39–144]	120 [40–100]	27 [40–100]
		PP		
Fixed Angles*	-	30 [39–144]	10 [100]	11 [40–100]
Fixed Angles†	-	30 [39–144]	10 [100]	11 [40–100]
Fourier*	20 [40–50]	31 [39–144]	120 [40–100]	28 [40–100]
Interp.*	20 [40–50]	31 [39–144]	120 [40–100]	28 [40–100]
Linear Ramp*	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp†	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Recursive TS	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–100]
TQA*	20 [40–50]	20 [39–144]	120 [40–100]	27 [40–100]
TQA	20 [40–50]	20 [39–144]	120 [40–100]	27 [40–100]
		SV		
Fixed Angles*	17 [11–20]	-	19 [10–20]	50 [10–20]
Fixed Angles†	17 [11–20]	-	19 [10–20]	50 [10–20]
Fourier*	92 [10–20]	20 [12–21]	778 [10–20]	382 [10–20]
Interp.*	92 [10–20]	20 [12–21]	779 [10–20]	382 [10–20]
Linear Ramp*	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp†	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS	92 [10–20]	20 [12–21]	407 [10–20]	379 [10–20]
TQA*	90 [10–20]	20 [12–21]	672 [10–20]	142 [10–20]
TQA	92 [10–20]	20 [12–21]	677 [10–20]	356 [10–20]

Table 22: **Number of Instances (num. nodes in brackets) for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angles*	-	92.5 ± 0.9	94.0 ± 1.4	97.2 ± 3.9
Fixed Angles†	-	90.8 ± 1.8	90.2 ± 2.9	96.8 ± 4.3
Fourier*	77.9 ± 5.5	85.2 ± 3.2	82.5 ± 8.6	81.8 ± 9.6
Interp.*	94.2 ± 2.1	90.0 ± 4.7	94.7 ± 2.6	86.6 ± 6.8
Linear Ramp*	94.5 ± 1.8	84.2 ± 1.4	95.6 ± 0.7	89.2 ± 2.4
Linear Ramp	92.4 ± 1.2	83.9 ± 2.2	94.0 ± 1.0	87.7 ± 2.1
Linear Ramp†	85.0 ± 3.1	77.6 ± 1.4	91.0 ± 2.8	65.8 ± 4.0
Recursive TS	93.0 ± 5.0	84.4 ± 4.7	93.5 ± 2.4	91.4 ± 7.4
TQA*	85.8 ± 2.2	86.5 ± 3.2	88.7 ± 5.9	92.2 ± 6.2
TQA	81.5 ± 3.5	85.3 ± 2.7	83.8 ± 7.1	87.3 ± 6.6
MPS (Quimb)				
Fixed Angles*	-	56.5 ± 3.5	96.7 ± 0.8	97.9 ± 1.3
Fixed Angles†	-	51.9 ± 2.6	96.6 ± 0.8	96.9 ± 3.3
Fourier*	81.3 ± 6.6	76.8 ± 10.2	88.2 ± 5.2	91.1 ± 8.2
Interp.*	96.9 ± 1.3	84.3 ± 10.1	95.6 ± 1.0	93.9 ± 6.5
Linear Ramp*	96.7 ± 0.8	69.0 ± 2.7	94.9 ± 1.2	93.8 ± 0.8
Linear Ramp	95.3 ± 1.2	69.4 ± 1.9	93.4 ± 1.2	90.6 ± 3.3
Linear Ramp†	93.1 ± 1.5	66.4 ± 1.2	90.5 ± 2.9	66.0 ± 5.0
Recursive TS	96.9 ± 0.8	70.4 ± 13.7	93.3 ± 1.1	89.9 ± 5.6
TQA*	91.1 ± 3.7	72.6 ± 18.2	92.2 ± 1.3	93.4 ± 6.7
TQA	82.2 ± 6.9	71.3 ± 17.4	89.3 ± 2.0	90.1 ± 8.0
PP				
Fixed Angles*	-	93.4 ± 0.6	90.3 ± 1.9	93.2 ± 1.6
Fixed Angles†	-	91.0 ± 1.4	86.2 ± 1.7	92.6 ± 1.7
Fourier*	91.0 ± 1.9	90.1 ± 3.0	85.7 ± 2.0	90.1 ± 2.1
Interp.*	93.7 ± 2.0	93.9 ± 0.9	88.8 ± 1.8	91.5 ± 2.2
Linear Ramp*	95.5 ± 1.0	92.8 ± 0.6	90.3 ± 2.2	91.0 ± 0.8
Linear Ramp	94.1 ± 1.0	91.6 ± 0.6	89.0 ± 2.3	90.4 ± 0.8
Linear Ramp†	93.7 ± 1.0	83.3 ± 1.1	86.7 ± 1.1	87.8 ± 2.4
Recursive TS	90.2 ± 1.3	89.0 ± 1.4	86.7 ± 2.1	88.3 ± 1.7
TQA*	91.7 ± 2.3	90.8 ± 0.7	85.6 ± 2.5	91.5 ± 1.5
TQA	84.6 ± 2.2	88.2 ± 1.0	77.4 ± 6.7	89.1 ± 2.4
SV				
Fixed Angles*	96.3 ± 1.4	-	98.0 ± 1.0	97.8 ± 1.1
Fixed Angles†	91.5 ± 3.3	-	96.7 ± 1.1	97.2 ± 1.1
Fourier*	78.9 ± 12.3	76.0 ± 10.2	79.7 ± 10.9	83.5 ± 13.1
Interp.*	95.7 ± 1.2	94.3 ± 1.1	95.2 ± 3.8	93.5 ± 5.5
Linear Ramp*	95.0 ± 0.9	94.3 ± 0.6	94.4 ± 1.5	95.5 ± 0.7
Linear Ramp	93.2 ± 1.2	93.1 ± 0.7	92.0 ± 1.7	93.8 ± 1.7
Linear Ramp†	89.6 ± 1.5	82.8 ± 3.9	89.3 ± 1.9	90.8 ± 3.1
Recursive TS	94.4 ± 1.8	92.2 ± 2.4	93.0 ± 4.9	93.5 ± 3.2
TQA*	94.4 ± 1.3	91.9 ± 1.7	93.1 ± 3.3	95.6 ± 1.7
TQA	90.3 ± 3.4	88.0 ± 3.6	83.6 ± 8.5	90.0 ± 4.9

Table 23: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for MAXCUT for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	R
				MPS (Aer)				
Fixed Angles⋆	-	92.5 ± 0.9	94.0 ± 1.4	97.2 ± 3.9	-	10 [39]	11 [100]	11 [40
Fixed Angles†	-	90.8 ± 1.8	90.2 ± 2.9	96.8 ± 4.3	-	10 [39]	11 [100]	11 [40
Fourier⋆	77.9 ± 5.5	85.2 ± 3.2	82.5 ± 8.6	81.8 ± 9.6	20 [40–50]	20 [39–144]	50 [40–100]	17 [40
Interp.⋆	94.2 ± 2.1	90.0 ± 4.7	94.7 ± 2.6	86.6 ± 6.8	20 [40–50]	20 [39–144]	50 [40–100]	17 [40
Linear Ramp⋆	94.5 ± 1.8	84.2 ± 1.4	95.6 ± 0.7	89.2 ± 2.4	10 [50]	10 [144]	10 [100]	7 [1
Linear Ramp	92.4 ± 1.2	83.9 ± 2.2	94.0 ± 1.0	87.7 ± 2.1	10 [50]	10 [144]	10 [100]	7 [1
Linear Ramp†	85.0 ± 3.1	77.6 ± 1.4	91.0 ± 2.8	65.8 ± 4.0	10 [50]	10 [144]	10 [100]	7 [1
Recursive TS	93.0 ± 5.0	84.4 ± 4.7	93.5 ± 2.4	91.4 ± 7.4	20 [40–50]	20 [39–144]	50 [40–100]	11 [40
TQA⋆	85.8 ± 2.2	86.5 ± 3.2	88.7 ± 5.9	92.2 ± 6.2	20 [40–50]	20 [39–144]	50 [40–100]	17 [40
TQA	81.5 ± 3.5	85.3 ± 2.7	83.8 ± 7.1	87.3 ± 6.6	20 [40–50]	20 [39–144]	50 [40–100]	17 [40
				MPS (Quimb)				
Fixed Angles⋆	-	56.5 ± 3.5	96.7 ± 0.8	97.9 ± 1.3	-	10 [144]	10 [100]	11 [40
Fixed Angles†	-	51.9 ± 2.6	96.6 ± 0.8	96.9 ± 3.3	-	10 [144]	10 [100]	11 [40
Fourier⋆	81.3 ± 6.6	76.8 ± 10.2	88.2 ± 5.2	91.1 ± 8.2	20 [40–50]	31 [39–144]	120 [40–100]	28 [40
Interp.⋆	96.9 ± 1.3	84.3 ± 10.1	95.6 ± 1.0	93.9 ± 6.5	20 [40–50]	31 [39–144]	120 [40–100]	28 [40
Linear Ramp⋆	96.7 ± 0.8	69.0 ± 2.7	94.9 ± 1.2	93.8 ± 0.8	10 [50]	10 [144]	10 [100]	7 [1
Linear Ramp	95.3 ± 1.2	69.4 ± 1.9	93.4 ± 1.2	90.6 ± 3.3	10 [50]	10 [144]	10 [100]	7 [1
Linear Ramp†	93.1 ± 1.5	66.4 ± 1.2	90.5 ± 2.9	66.0 ± 5.0	10 [50]	10 [144]	10 [100]	7 [1
Recursive TS	96.9 ± 0.8	70.4 ± 13.7	93.3 ± 1.1	89.9 ± 5.6	20 [40–50]	31 [39–144]	10 [100]	18 [40
TQA⋆	91.1 ± 3.7	72.6 ± 18.2	92.2 ± 1.3	93.4 ± 6.7	20 [40–50]	20 [39–144]	120 [40–100]	27 [40
TQA	82.2 ± 6.9	71.3 ± 17.4	89.3 ± 2.0	90.1 ± 8.0	20 [40–50]	20 [39–144]	120 [40–100]	27 [40
				PP				
Fixed Angles⋆	-	93.4 ± 0.6	90.3 ± 1.9	93.2 ± 1.6	-	30 [39–144]	10 [100]	11 [40
Fixed Angles†	-	91.0 ± 1.4	86.2 ± 1.7	92.6 ± 1.7	-	30 [39–144]	10 [100]	11 [40
Fourier⋆	91.0 ± 1.9	90.1 ± 3.0	85.7 ± 2.0	90.1 ± 2.1	20 [40–50]	31 [39–144]	120 [40–100]	28 [40
Interp.⋆	93.7 ± 2.0	93.9 ± 0.9	88.8 ± 1.8	91.5 ± 2.2	20 [40–50]	31 [39–144]	120 [40–100]	28 [40
Linear Ramp⋆	95.5 ± 1.0	92.8 ± 0.6	90.3 ± 2.2	91.0 ± 0.8	10 [50]	20 [105–144]	20 [40–100]	8 [1
Linear Ramp	94.1 ± 1.0	91.6 ± 0.6	89.0 ± 2.3	90.4 ± 0.8	10 [50]	20 [105–144]	20 [40–100]	8 [1
Linear Ramp†	93.7 ± 1.0	83.3 ± 1.1	86.7 ± 1.1	87.8 ± 2.4	10 [50]	20 [105–144]	20 [40–100]	8 [1
Recursive TS	90.2 ± 1.3	89.0 ± 1.4	86.7 ± 2.1	88.3 ± 1.7	20 [40–50]	30 [39–144]	111 [40–100]	18 [40
TQA⋆	91.7 ± 2.3	90.8 ± 0.7	85.6 ± 2.5	91.5 ± 1.5	20 [40–50]	20 [39–144]	120 [40–100]	27 [40
TQA	84.6 ± 2.2	88.2 ± 1.0	77.4 ± 6.7	89.1 ± 2.4	20 [40–50]	20 [39–144]	120 [40–100]	27 [40
				SV				
Fixed Angles⋆	96.3 ± 1.4	-	98.0 ± 1.0	97.8 ± 1.1	17 [11–20]	-	19 [10–20]	50 [1
Fixed Angles†	91.5 ± 3.3	-	96.7 ± 1.1	97.2 ± 1.1	17 [11–20]	-	19 [10–20]	50 [1
Fourier⋆	78.9 ± 12.3	76.0 ± 10.2	79.7 ± 10.9	83.5 ± 13.1	92 [10–20]	20 [12–21]	778 [10–20]	382 [1
Interp.⋆	95.7 ± 1.2	94.3 ± 1.1	95.2 ± 3.8	93.5 ± 5.5	92 [10–20]	20 [12–21]	779 [10–20]	382 [1
Linear Ramp⋆	95.0 ± 0.9	94.3 ± 0.6	94.4 ± 1.5	95.5 ± 0.7	10 [20]	10 [21]	10 [20]	7 [
Linear Ramp	93.2 ± 1.2	93.1 ± 0.7	92.0 ± 1.7	93.8 ± 1.7	10 [20]	10 [21]	10 [20]	7 [
Linear Ramp†	89.6 ± 1.5	82.8 ± 3.9	89.3 ± 1.9	90.8 ± 3.1	10 [20]	10 [21]	10 [20]	7 [
Recursive TS	94.4 ± 1.8	92.2 ± 2.4	93.0 ± 4.9	93.5 ± 3.2	92 [10–20]	20 [12–21]	407 [10–20]	379 [1
TQA⋆	94.4 ± 1.3	91.9 ± 1.7	93.1 ± 3.3	95.6 ± 1.7	90 [10–20]	20 [12–21]	672 [10–20]	142 [1
TQA	90.3 ± 3.4	88.0 ± 3.6	83.6 ± 8.5	90.0 ± 4.9	92 [10–20]	20 [12–21]	677 [10–20]	356 [1

Table 24: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

9 Tables for depth $P = 9$

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angles⋆	-	-	1 [100]	1 [100]
Fixed Angles†	-	-	1 [100]	1 [100]
Fourier⋆	20 [40–50]	20 [39–144]	50 [40–100]	17 [40–100]
Interp.⋆	20 [40–50]	20 [39–144]	50 [40–100]	17 [40–100]
Linear Ramp⋆	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
TQA⋆	10 [50]	10 [144]	10 [100]	7 [100]
TQA	10 [50]	10 [144]	10 [100]	7 [100]
MPS (Quimb)				
Fixed Angles⋆	-	10 [144]	1 [100]	1 [100]
Fixed Angles†	-	10 [144]	1 [100]	1 [100]
Fourier⋆	20 [40–50]	31 [39–144]	111 [40–100]	18 [40–100]
Interp.⋆	20 [40–50]	31 [39–144]	111 [40–100]	18 [40–100]
Linear Ramp⋆	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS	20 [40–50]	31 [39–144]	10 [100]	18 [40–100]
TQA⋆	10 [50]	10 [144]	10 [100]	7 [100]
TQA	10 [50]	10 [144]	10 [100]	7 [100]
PP				
Fixed Angles⋆	-	10 [144]	1 [100]	1 [100]
Fixed Angles†	-	10 [144]	1 [100]	1 [100]
Fourier⋆	20 [40–50]	31 [39–144]	111 [40–100]	18 [40–100]
Interp.⋆	20 [40–50]	31 [39–144]	111 [40–100]	18 [40–100]
Linear Ramp⋆	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp†	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Recursive TS	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–100]
TQA⋆	10 [50]	10 [144]	10 [100]	7 [100]
TQA	10 [50]	10 [144]	10 [100]	7 [100]
SV				
Fixed Angles⋆	17 [11–20]	-	19 [10–20]	50 [10–20]
Fixed Angles†	17 [11–20]	-	19 [10–20]	50 [10–20]
Fourier⋆	92 [10–20]	20 [12–21]	776 [10–20]	380 [10–20]
Interp.⋆	92 [10–20]	20 [12–21]	777 [10–20]	381 [10–20]
Linear Ramp⋆	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp†	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS	92 [10–20]	20 [12–21]	407 [10–20]	376 [10–20]
TQA⋆	90 [10–20]	20 [12–21]	671 [10–20]	141 [10–20]
TQA	92 [10–20]	20 [12–21]	676 [10–20]	356 [10–20]

Table 25: **Number of Instances (num. nodes in brackets) for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angles*	-	-	97.8	90.2
Fixed Angles†	-	-	97.5	85.1
Fourier*	78.1 ± 5.3	85.8 ± 4.0	79.5 ± 8.3	84.0 ± 9.3
Interp.*	94.3 ± 3.6	90.6 ± 4.8	94.8 ± 2.5	88.7 ± 7.1
Linear Ramp*	95.4 ± 1.8	84.2 ± 2.0	96.0 ± 0.7	91.3 ± 2.1
Linear Ramp	93.1 ± 1.4	84.4 ± 3.0	94.6 ± 1.0	88.6 ± 3.0
Linear Ramp†	87.4 ± 2.6	78.4 ± 1.8	91.9 ± 2.6	66.8 ± 4.9
Recursive TS	94.7 ± 2.5	84.6 ± 5.1	93.8 ± 2.5	92.4 ± 6.2
TQA*	86.5 ± 1.5	84.2 ± 1.9	93.8 ± 1.4	88.5 ± 3.9
TQA	85.4 ± 1.5	83.6 ± 1.7	91.3 ± 2.1	83.8 ± 7.3
MPS (Quimb)				
Fixed Angles*	-	59.1 ± 3.3	97.8	96.1
Fixed Angles†	-	52.4 ± 3.3	97.4	90.9
Fourier*	82.0 ± 7.8	78.3 ± 9.7	88.6 ± 4.3	88.8 ± 8.8
Interp.*	97.2 ± 1.4	86.2 ± 9.3	96.2 ± 0.9	90.9 ± 7.3
Linear Ramp*	97.0 ± 1.1	72.1 ± 1.9	95.4 ± 1.0	92.9 ± 3.0
Linear Ramp	95.5 ± 1.3	70.4 ± 1.5	94.1 ± 1.1	91.5 ± 2.0
Linear Ramp†	94.1 ± 1.5	67.1 ± 1.1	91.4 ± 2.7	66.9 ± 5.9
Recursive TS	97.2 ± 0.9	71.0 ± 13.3	93.6 ± 1.3	90.8 ± 5.1
TQA*	90.0 ± 2.3	57.8 ± 1.2	93.3 ± 1.0	90.1 ± 1.5
TQA	83.7 ± 7.3	57.4 ± 1.1	90.5 ± 1.8	81.6 ± 12.0
PP				
Fixed Angles*	-	93.7 ± 0.4	91.5	91.2
Fixed Angles†	-	90.9 ± 1.5	87.2	90.7
Fourier*	91.7 ± 1.6	90.6 ± 3.2	86.2 ± 1.9	90.5 ± 2.3
Interp.*	94.1 ± 2.0	94.4 ± 0.8	89.2 ± 1.7	90.8 ± 2.3
Linear Ramp*	95.6 ± 1.0	93.3 ± 0.5	90.7 ± 2.2	91.4 ± 0.8
Linear Ramp	94.4 ± 1.0	92.2 ± 0.6	89.5 ± 2.2	90.8 ± 0.8
Linear Ramp†	93.9 ± 1.0	84.3 ± 1.1	87.5 ± 1.2	88.5 ± 2.1
Recursive TS	90.3 ± 1.3	89.4 ± 1.5	87.0 ± 2.2	88.4 ± 1.7
TQA*	92.9 ± 1.4	91.0 ± 0.6	85.6 ± 2.9	90.4 ± 0.9
TQA	83.6 ± 0.9	89.1 ± 0.7	77.1 ± 7.9	86.9 ± 3.0
SV				
Fixed Angles*	96.9 ± 1.4	-	98.4 ± 0.8	98.3 ± 1.1
Fixed Angles†	91.7 ± 3.5	-	97.0 ± 1.3	97.7 ± 1.0
Fourier*	78.5 ± 12.6	73.8 ± 10.4	78.9 ± 11.5	83.1 ± 13.8
Interp.*	96.2 ± 1.1	94.5 ± 1.1	95.5 ± 4.1	93.4 ± 6.2
Linear Ramp*	95.4 ± 0.8	94.7 ± 0.6	94.8 ± 1.5	95.9 ± 0.9
Linear Ramp	93.7 ± 1.2	93.8 ± 0.5	92.8 ± 1.5	94.3 ± 1.5
Linear Ramp†	90.4 ± 1.4	83.8 ± 3.9	90.2 ± 1.8	91.5 ± 2.9
Recursive TS	94.8 ± 1.8	92.5 ± 2.4	93.1 ± 4.4	93.8 ± 3.1
TQA*	94.9 ± 1.3	92.5 ± 1.6	93.6 ± 3.3	96.1 ± 1.7
TQA	91.0 ± 3.6	88.9 ± 3.4	83.8 ± 9.2	90.8 ± 4.9

Table 26: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for MAXCUT for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	R
MPS (Aer)								
Fixed Angles⋆	-	-	97.8	90.2	-	-	1 [100]	1 [1
Fixed Angles†	-	-	97.5	85.1	-	-	1 [100]	1 [1
Fourier⋆	78.1 ± 5.3	85.8 ± 4.0	79.5 ± 8.3	84.0 ± 9.3	20 [40–50]	20 [39–144]	50 [40–100]	17 [40
Interp.⋆	94.3 ± 3.6	90.6 ± 4.8	94.8 ± 2.5	88.7 ± 7.1	20 [40–50]	20 [39–144]	50 [40–100]	17 [40
Linear Ramp⋆	95.4 ± 1.8	84.2 ± 2.0	96.0 ± 0.7	91.3 ± 2.1	10 [50]	10 [144]	10 [100]	7 [1
Linear Ramp	93.1 ± 1.4	84.4 ± 3.0	94.6 ± 1.0	88.6 ± 3.0	10 [50]	10 [144]	10 [100]	7 [1
Linear Ramp†	87.4 ± 2.6	78.4 ± 1.8	91.9 ± 2.6	66.8 ± 4.9	10 [50]	10 [144]	10 [100]	7 [1
Recursive TS	94.7 ± 2.5	84.6 ± 5.1	93.8 ± 2.5	92.4 ± 6.2	20 [40–50]	20 [39–144]	50 [40–100]	11 [40
TQA⋆	86.5 ± 1.5	84.2 ± 1.9	93.8 ± 1.4	88.5 ± 3.9	10 [50]	10 [144]	10 [100]	7 [1
TQA	85.4 ± 1.5	83.6 ± 1.7	91.3 ± 2.1	83.8 ± 7.3	10 [50]	10 [144]	10 [100]	7 [1
MPS (Quimb)								
Fixed Angles⋆	-	59.1 ± 3.3	97.8	96.1	-	10 [144]	1 [100]	1 [1
Fixed Angles†	-	52.4 ± 3.3	97.4	90.9	-	10 [144]	1 [100]	1 [1
Fourier⋆	82.0 ± 7.8	78.3 ± 9.7	88.6 ± 4.3	88.8 ± 8.8	20 [40–50]	31 [39–144]	111 [40–100]	18 [40
Interp.⋆	97.2 ± 1.4	86.2 ± 9.3	96.2 ± 0.9	90.9 ± 7.3	20 [40–50]	31 [39–144]	111 [40–100]	18 [40
Linear Ramp⋆	97.0 ± 1.1	72.1 ± 1.9	95.4 ± 1.0	92.9 ± 3.0	10 [50]	10 [144]	10 [100]	7 [1
Linear Ramp	95.5 ± 1.3	70.4 ± 1.5	94.1 ± 1.1	91.5 ± 2.0	10 [50]	10 [144]	10 [100]	7 [1
Linear Ramp†	94.1 ± 1.5	67.1 ± 1.1	91.4 ± 2.7	66.9 ± 5.9	10 [50]	10 [144]	10 [100]	7 [1
Recursive TS	97.2 ± 0.9	71.0 ± 13.3	93.6 ± 1.3	90.8 ± 5.1	20 [40–50]	31 [39–144]	10 [100]	18 [40
TQA⋆	90.0 ± 2.3	57.8 ± 1.2	93.3 ± 1.0	90.1 ± 1.5	10 [50]	10 [144]	10 [100]	7 [1
TQA	83.7 ± 7.3	57.4 ± 1.1	90.5 ± 1.8	81.6 ± 12.0	10 [50]	10 [144]	10 [100]	7 [1
PP								
Fixed Angles⋆	-	93.7 ± 0.4	91.5	91.2	-	10 [144]	1 [100]	1 [1
Fixed Angles†	-	90.9 ± 1.5	87.2	90.7	-	10 [144]	1 [100]	1 [1
Fourier⋆	91.7 ± 1.6	90.6 ± 3.2	86.2 ± 1.9	90.5 ± 2.3	20 [40–50]	31 [39–144]	111 [40–100]	18 [40
Interp.⋆	94.1 ± 2.0	94.4 ± 0.8	89.2 ± 1.7	90.8 ± 2.3	20 [40–50]	31 [39–144]	111 [40–100]	18 [40
Linear Ramp⋆	95.6 ± 1.0	93.3 ± 0.5	90.7 ± 2.2	91.4 ± 0.8	10 [50]	20 [105–144]	20 [40–100]	8 [1
Linear Ramp	94.4 ± 1.0	92.2 ± 0.6	89.5 ± 2.2	90.8 ± 0.8	10 [50]	20 [105–144]	20 [40–100]	8 [1
Linear Ramp†	93.9 ± 1.0	84.3 ± 1.1	87.5 ± 1.2	88.5 ± 2.1	10 [50]	20 [105–144]	20 [40–100]	8 [1
Recursive TS	90.3 ± 1.3	89.4 ± 1.5	87.0 ± 2.2	88.4 ± 1.7	20 [40–50]	30 [39–144]	111 [40–100]	18 [40
TQA⋆	92.9 ± 1.4	91.0 ± 0.6	85.6 ± 2.9	90.4 ± 0.9	10 [50]	10 [144]	10 [100]	7 [1
TQA	83.6 ± 0.9	89.1 ± 0.7	77.1 ± 7.9	86.9 ± 3.0	10 [50]	10 [144]	10 [100]	7 [1
SV								
Fixed Angles⋆	96.9 ± 1.4	-	98.4 ± 0.8	98.3 ± 1.1	17 [11–20]	-	19 [10–20]	50 [1
Fixed Angles†	91.7 ± 3.5	-	97.0 ± 1.3	97.7 ± 1.0	17 [11–20]	-	19 [10–20]	50 [1
Fourier⋆	78.5 ± 12.6	73.8 ± 10.4	78.9 ± 11.5	83.1 ± 13.8	92 [10–20]	20 [12–21]	776 [10–20]	380 [1
Interp.⋆	96.2 ± 1.1	94.5 ± 1.1	95.5 ± 4.1	93.4 ± 6.2	92 [10–20]	20 [12–21]	777 [10–20]	381 [1
Linear Ramp⋆	95.4 ± 0.8	94.7 ± 0.6	94.8 ± 1.5	95.9 ± 0.9	10 [20]	10 [21]	10 [20]	7 [
Linear Ramp	93.7 ± 1.2	93.8 ± 0.5	92.8 ± 1.5	94.3 ± 1.5	10 [20]	10 [21]	10 [20]	7 [
Linear Ramp†	90.4 ± 1.4	83.8 ± 3.9	90.2 ± 1.8	91.5 ± 2.9	10 [20]	10 [21]	10 [20]	7 [
Recursive TS	94.8 ± 1.8	92.5 ± 2.4	93.1 ± 4.4	93.8 ± 3.1	92 [10–20]	20 [12–21]	407 [10–20]	376 [1
TQA⋆	94.9 ± 1.3	92.5 ± 1.6	93.6 ± 3.3	96.1 ± 1.7	90 [10–20]	20 [12–21]	671 [10–20]	141 [1
TQA	91.0 ± 3.6	88.9 ± 3.4	83.8 ± 9.2	90.8 ± 4.9	92 [10–20]	20 [12–21]	676 [10–20]	356 [1

Table 27: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

10 Tables for depth $P = 10$

	ER	HH	LB	RR
		MPS (Aer)		
Fixed Angles*	-	20 [39–144]	14 [100]	11 [40–100]
Fixed Angles†	-	20 [39–144]	14 [100]	11 [40–100]
Fourier*	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
Interp.*	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
Linear Ramp*	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
Linear Ramp	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
Linear Ramp†	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
Recursive TS	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
TQA*	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
TQA	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
		MPS (Quimb)		
Fixed Angles*	-	-	1 [100]	12 [40–100]
Fixed Angles†	-	-	1 [100]	12 [40–100]
Fourier*	20 [40–50]	31 [39–144]	10 [100]	12 [40–100]
Interp.*	20 [40–50]	31 [39–144]	10 [100]	12 [40–100]
Linear Ramp*	20 [40–50]	31 [39–144]	10 [100]	12 [40–100]
Linear Ramp	20 [40–50]	31 [39–144]	10 [100]	12 [40–100]
Linear Ramp†	20 [40–50]	31 [39–144]	10 [100]	12 [40–100]
Recursive TS	20 [40–50]	31 [39–144]	10 [100]	12 [40–100]
TQA*	20 [40–50]	31 [39–144]	10 [100]	12 [40–100]
TQA	20 [40–50]	31 [39–144]	10 [100]	12 [40–100]
		PP		
Fixed Angles*	-	30 [39–144]	10 [100]	12 [40–100]
Fixed Angles†	-	30 [39–144]	10 [100]	12 [40–100]
Fourier*	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–100]
Interp.*	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–100]
Linear Ramp*	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–100]
Linear Ramp	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–100]
Linear Ramp†	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–100]
Recursive TS	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–100]
TQA*	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–100]
TQA	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–100]
		SV		
Fixed Angles*	17 [11–20]	-	16 [10–20]	60 [10–20]
Fixed Angles†	17 [11–20]	-	16 [10–20]	60 [10–20]
Fourier*	92 [10–20]	20 [12–21]	180 [10–20]	184 [10–20]
Interp.*	92 [10–20]	20 [12–21]	180 [10–20]	184 [10–20]
Linear Ramp*	92 [10–20]	20 [12–21]	180 [10–20]	184 [10–20]
Linear Ramp	92 [10–20]	20 [12–21]	180 [10–20]	184 [10–20]
Linear Ramp†	92 [10–20]	20 [12–21]	180 [10–20]	184 [10–20]
Recursive TS	92 [10–20]	20 [12–21]	180 [10–20]	184 [10–20]
TQA*	92 [10–20]	20 [12–21]	180 [10–20]	184 [10–20]
TQA	92 [10–20]	20 [12–21]	180 [10–20]	184 [10–20]

Table 28: **Number of Instances (num. nodes in brackets) for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angles*	-	89.7 ± 3.9	96.4 ± 1.0	98.5 ± 1.4
Fixed Angles†	-	88.3 ± 3.6	94.5 ± 2.3	97.8 ± 3.3
Fourier*	81.2 ± 6.1	85.1 ± 4.8	80.8 ± 8.9	90.1 ± 5.9
Interp.*	94.2 ± 4.0	91.3 ± 4.6	95.5 ± 2.4	91.1 ± 3.3
Linear Ramp*	96.5 ± 0.8	89.2 ± 4.9	95.5 ± 1.7	97.4 ± 2.2
Linear Ramp	94.4 ± 1.8	92.1 ± 1.8	94.3 ± 2.0	96.0 ± 2.0
Linear Ramp†	90.1 ± 2.1	83.3 ± 1.5	91.7 ± 3.8	81.2 ± 7.5
Recursive TS	95.8 ± 1.8	85.0 ± 5.1	94.1 ± 2.5	93.9 ± 4.6
TQA*	89.4 ± 2.0	87.9 ± 2.9	89.8 ± 3.0	95.5 ± 3.5
TQA	86.5 ± 2.1	87.1 ± 2.5	85.4 ± 4.4	92.7 ± 4.6
MPS (Quimb)				
Fixed Angles*	-	-	98.1	98.4 ± 1.2
Fixed Angles†	-	-	97.7	97.7 ± 2.2
Fourier*	79.6 ± 6.1	77.4 ± 10.9	88.3 ± 5.6	91.8 ± 9.3
Interp.*	97.4 ± 1.5	86.6 ± 10.1	97.1 ± 0.6	92.9 ± 5.3
Linear Ramp*	97.5 ± 0.7	82.7 ± 10.3	95.9 ± 1.0	96.8 ± 2.1
Linear Ramp	96.4 ± 0.9	82.6 ± 9.6	94.6 ± 1.1	94.7 ± 3.1
Linear Ramp†	95.0 ± 1.3	75.6 ± 7.6	92.2 ± 2.5	82.1 ± 10.8
Recursive TS	97.4 ± 0.9	70.8 ± 13.9	93.8 ± 1.4	94.0 ± 2.6
TQA*	91.8 ± 2.9	80.8 ± 9.3	92.9 ± 1.8	96.1 ± 2.2
TQA	87.7 ± 1.8	79.6 ± 9.2	90.5 ± 2.6	92.8 ± 5.2
PP				
Fixed Angles*	-	94.2 ± 0.6	90.7 ± 1.9	93.5 ± 1.6
Fixed Angles†	-	91.5 ± 1.3	86.7 ± 1.7	92.9 ± 1.6
Fourier*	91.5 ± 1.9	91.3 ± 3.1	86.0 ± 2.2	90.7 ± 2.4
Interp.*	94.4 ± 1.9	95.0 ± 0.7	89.6 ± 1.7	90.4 ± 2.5
Linear Ramp*	95.6 ± 1.2	93.5 ± 0.6	90.6 ± 2.5	92.7 ± 1.5
Linear Ramp	94.7 ± 1.2	93.0 ± 0.7	89.4 ± 2.4	91.9 ± 1.5
Linear Ramp†	93.9 ± 1.1	85.2 ± 1.2	87.7 ± 2.1	88.1 ± 1.9
Recursive TS	90.4 ± 1.2	89.6 ± 1.5	87.2 ± 2.2	88.6 ± 1.9
TQA*	91.3 ± 2.5	91.8 ± 0.7	85.6 ± 3.0	91.6 ± 1.6
TQA	84.4 ± 2.4	90.0 ± 0.9	77.5 ± 7.8	89.4 ± 3.0
SV				
Fixed Angles*	97.4 ± 1.4	-	97.5 ± 1.3	98.7 ± 1.0
Fixed Angles†	92.0 ± 3.6	-	96.3 ± 1.6	98.0 ± 1.0
Fourier*	78.1 ± 13.1	74.2 ± 12.3	76.8 ± 11.3	80.7 ± 14.0
Interp.*	96.5 ± 1.1	94.7 ± 1.0	95.4 ± 1.9	92.1 ± 6.8
Linear Ramp*	96.7 ± 1.1	95.3 ± 0.8	95.5 ± 1.6	97.8 ± 1.2
Linear Ramp	95.0 ± 1.4	94.6 ± 0.9	93.5 ± 1.9	96.2 ± 1.7
Linear Ramp†	91.3 ± 2.1	84.8 ± 4.3	90.5 ± 3.2	92.8 ± 3.3
Recursive TS	95.0 ± 2.0	92.8 ± 2.4	92.5 ± 2.2	93.8 ± 3.1
TQA*	95.3 ± 1.3	92.8 ± 1.9	93.2 ± 2.5	96.7 ± 1.5
TQA	91.4 ± 3.7	89.8 ± 3.3	84.1 ± 8.2	91.9 ± 4.6

Table 29: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for MAXCUT for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	RR
MPS (Aer)								
Fixed Angles*	-	89.7 ± 3.9	96.4 ± 1.0	98.5 ± 1.4	-	20 [39–144]	14 [100]	11 [40-
Fixed Angles†	-	88.3 ± 3.6	94.5 ± 2.3	97.8 ± 3.3	-	20 [39–144]	14 [100]	11 [40-
Fourier*	81.2 ± 6.1	85.1 ± 4.8	80.8 ± 8.9	90.1 ± 5.9	20 [40–50]	20 [39–144]	50 [40–100]	11 [40-
Interp.*	94.2 ± 4.0	91.3 ± 4.6	95.5 ± 2.4	91.1 ± 3.3	20 [40–50]	20 [39–144]	50 [40–100]	11 [40-
Linear Ramp*	96.5 ± 0.8	89.2 ± 4.9	95.5 ± 1.7	97.4 ± 2.2	20 [40–50]	20 [39–144]	50 [40–100]	11 [40-
Linear Ramp	94.4 ± 1.8	92.1 ± 1.8	94.3 ± 2.0	96.0 ± 2.0	20 [40–50]	20 [39–144]	50 [40–100]	11 [40-
Linear Ramp†	90.1 ± 2.1	83.3 ± 1.5	91.7 ± 3.8	81.2 ± 7.5	20 [40–50]	20 [39–144]	50 [40–100]	11 [40-
Recursive TS	95.8 ± 1.8	85.0 ± 5.1	94.1 ± 2.5	93.9 ± 4.6	20 [40–50]	20 [39–144]	50 [40–100]	11 [40-
TQA*	89.4 ± 2.0	87.9 ± 2.9	89.8 ± 3.0	95.5 ± 3.5	20 [40–50]	20 [39–144]	50 [40–100]	11 [40-
TQA	86.5 ± 2.1	87.1 ± 2.5	85.4 ± 4.4	92.7 ± 4.6	20 [40–50]	20 [39–144]	50 [40–100]	11 [40-
MPS (Quimb)								
Fixed Angles*	-	-	98.1	98.4 ± 1.2	-	-	1 [100]	12 [40-
Fixed Angles†	-	-	97.7	97.7 ± 2.2	-	-	1 [100]	12 [40-
Fourier*	79.6 ± 6.1	77.4 ± 10.9	88.3 ± 5.6	91.8 ± 9.3	20 [40–50]	31 [39–144]	10 [100]	12 [40-
Interp.*	97.4 ± 1.5	86.6 ± 10.1	97.1 ± 0.6	92.9 ± 5.3	20 [40–50]	31 [39–144]	10 [100]	12 [40-
Linear Ramp*	97.5 ± 0.7	82.7 ± 10.3	95.9 ± 1.0	96.8 ± 2.1	20 [40–50]	31 [39–144]	10 [100]	12 [40-
Linear Ramp	96.4 ± 0.9	82.6 ± 9.6	94.6 ± 1.1	94.7 ± 3.1	20 [40–50]	31 [39–144]	10 [100]	12 [40-
Linear Ramp†	95.0 ± 1.3	75.6 ± 7.6	92.2 ± 2.5	82.1 ± 10.8	20 [40–50]	31 [39–144]	10 [100]	12 [40-
Recursive TS	97.4 ± 0.9	70.8 ± 13.9	93.8 ± 1.4	94.0 ± 2.6	20 [40–50]	31 [39–144]	10 [100]	12 [40-
TQA*	91.8 ± 2.9	80.8 ± 9.3	92.9 ± 1.8	96.1 ± 2.2	20 [40–50]	31 [39–144]	10 [100]	12 [40-
TQA	87.7 ± 1.8	79.6 ± 9.2	90.5 ± 2.6	92.8 ± 5.2	20 [40–50]	31 [39–144]	10 [100]	12 [40-
PP								
Fixed Angles*	-	94.2 ± 0.6	90.7 ± 1.9	93.5 ± 1.6	-	30 [39–144]	10 [100]	12 [40-
Fixed Angles†	-	91.5 ± 1.3	86.7 ± 1.7	92.9 ± 1.6	-	30 [39–144]	10 [100]	12 [40-
Fourier*	91.5 ± 1.9	91.3 ± 3.1	86.0 ± 2.2	90.7 ± 2.4	20 [40–50]	30 [39–144]	111 [40–100]	18 [40-
Interp.*	94.4 ± 1.9	95.0 ± 0.7	89.6 ± 1.7	90.4 ± 2.5	20 [40–50]	30 [39–144]	111 [40–100]	18 [40-
Linear Ramp*	95.6 ± 1.2	93.5 ± 0.6	90.6 ± 2.5	92.7 ± 1.5	20 [40–50]	30 [39–144]	111 [40–100]	18 [40-
Linear Ramp	94.7 ± 1.2	93.0 ± 0.7	89.4 ± 2.4	91.9 ± 1.5	20 [40–50]	30 [39–144]	111 [40–100]	18 [40-
Linear Ramp†	93.9 ± 1.1	85.2 ± 1.2	87.7 ± 2.1	88.1 ± 1.9	20 [40–50]	30 [39–144]	111 [40–100]	18 [40-
Recursive TS	90.4 ± 1.2	89.6 ± 1.5	87.2 ± 2.2	88.6 ± 1.9	20 [40–50]	30 [39–144]	111 [40–100]	18 [40-
TQA*	91.3 ± 2.5	91.8 ± 0.7	85.6 ± 3.0	91.6 ± 1.6	20 [40–50]	30 [39–144]	111 [40–100]	18 [40-
TQA	84.4 ± 2.4	90.0 ± 0.9	77.5 ± 7.8	89.4 ± 3.0	20 [40–50]	30 [39–144]	111 [40–100]	18 [40-
SV								
Fixed Angles*	97.4 ± 1.4	-	97.5 ± 1.3	98.7 ± 1.0	17 [11–20]	-	16 [10–20]	60 [10
Fixed Angles†	92.0 ± 3.6	-	96.3 ± 1.6	98.0 ± 1.0	17 [11–20]	-	16 [10–20]	60 [10
Fourier*	78.1 ± 13.1	74.2 ± 12.3	76.8 ± 11.3	80.7 ± 14.0	92 [10–20]	20 [12–21]	180 [10–20]	184 [10
Interp.*	96.5 ± 1.1	94.7 ± 1.0	95.4 ± 1.9	92.1 ± 6.8	92 [10–20]	20 [12–21]	180 [10–20]	184 [10
Linear Ramp*	96.7 ± 1.1	95.3 ± 0.8	95.5 ± 1.6	97.8 ± 1.2	92 [10–20]	20 [12–21]	180 [10–20]	184 [10
Linear Ramp	95.0 ± 1.4	94.6 ± 0.9	93.5 ± 1.9	96.2 ± 1.7	92 [10–20]	20 [12–21]	180 [10–20]	184 [10
Linear Ramp†	91.3 ± 2.1	84.8 ± 4.3	90.5 ± 3.2	92.8 ± 3.3	92 [10–20]	20 [12–21]	180 [10–20]	184 [10
Recursive TS	95.0 ± 2.0	92.8 ± 2.4	92.5 ± 2.2	93.8 ± 3.1	92 [10–20]	20 [12–21]	180 [10–20]	184 [10
TQA*	95.3 ± 1.3	92.8 ± 1.9	93.2 ± 2.5	96.7 ± 1.5	92 [10–20]	20 [12–21]	180 [10–20]	184 [10
TQA	91.4 ± 3.7	89.8 ± 3.3	84.1 ± 8.2	91.9 ± 4.6	92 [10–20]	20 [12–21]	180 [10–20]	184 [10

Table 30: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

Summary Tables of `MIS` Data

11 Tables for depth $P = 1$

	ER	HH	LB	RR
	MPS (Aer)			
Fourier*	10 [70]	10 [144]	10 [100]	7 [100]
Interp.*	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	10 [70]	10 [144]	10 [100]	7 [100]
TQA*	-	-	-	-
TQA	-	-	-	-
	MPS (Quimb)			
Fourier*	10 [70]	10 [144]	10 [100]	7 [100]
Interp.*	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	10 [70]	10 [144]	10 [100]	7 [100]
TQA*	-	-	-	-
TQA	-	-	-	-
	PP			
Fourier*	10 [70]	10 [144]	10 [100]	7 [100]
Interp.*	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	10 [70]	10 [144]	10 [100]	7 [100]
TQA*	-	-	-	-
TQA	-	-	-	-
	SV			
Fourier*	10 [20]	10 [21]	10 [20]	7 [20]
Interp.*	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	10 [20]	10 [21]	10 [20]	7 [20]
TQA*	-	-	-	1 [20]
TQA	-	-	-	1 [20]

Table 31: **Number of Instances (num. nodes in brackets) for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
		MPS (Aer)		
Fourier*	56682.2 ± 2535.8	7484.9 ± 0.0	15317.6 ± 0.0	9231.0 ± 8.4
Interp.*	56598.1 ± 2542.4	7484.9 ± 0.0	15317.6 ± 0.0	9231.0 ± 8.4
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	7459.6 ± 2169.4	6888.2 ± 0.0	14411.5 ± 0.0	9179.9 ± 9.1
TQA*	-	-	-	-
TQA	-	-	-	-
		MPS (Quimb)		
Fourier*	56682.3 ± 2535.8	7484.9 ± 0.0	15317.6 ± 0.0	9231.0 ± 8.4
Interp.*	56608.9 ± 2547.5	7484.9 ± 0.0	15317.6 ± 0.0	9231.0 ± 8.4
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	7459.6 ± 2169.4	6888.2 ± 0.0	14411.5 ± 0.0	9179.9 ± 9.1
TQA*	-	-	-	-
TQA	-	-	-	-
		PP		
Fourier*	56682.2 ± 2535.8	7484.9 ± 0.0	15317.6 ± 0.0	9231.0 ± 8.4
Interp.*	56598.1 ± 2542.4	7484.9 ± 0.0	15317.6 ± 0.0	9231.0 ± 8.4
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	7459.6 ± 2169.4	6888.2 ± 0.0	14411.5 ± 0.0	9179.9 ± 9.1
TQA*	-	-	-	-
TQA	-	-	-	-
		SV		
Fourier*	2839.4 ± 631.4	940.5 ± 0.0	7569.7 ± 4127.4	5998.2 ± 3052.2
Interp.*	2839.4 ± 631.4	940.5 ± 0.0	7569.7 ± 4127.3	5998.2 ± 3052.2
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	2839.4 ± 631.4	940.5 ± 0.0	7569.7 ± 4127.3	5998.2 ± 3052.2
TQA*	-	-	-	1874.4
TQA	-	-	-	1806.3

Table 32: **Avg. Energy $\pm$ standard deviation for MIS for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	
MPS (Aer)								
Fourier⋆	56682.2 ± 2535.8	7484.9 ± 0.0	15317.6 ± 0.0	9231.0 ± 8.4	10 [70]	10 [144]	10 [100]	7
Interp.⋆	56598.1 ± 2542.4	7484.9 ± 0.0	15317.6 ± 0.0	9231.0 ± 8.4	10 [70]	10 [144]	10 [100]	7
Linear Ramp⋆	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS	7459.6 ± 2169.4	6888.2 ± 0.0	14411.5 ± 0.0	9179.9 ± 9.1	10 [70]	10 [144]	10 [100]	7
TQA⋆	-	-	-	-	-	-	-	
TQA	-	-	-	-	-	-	-	
MPS (Quimb)								
Fourier⋆	56682.3 ± 2535.8	7484.9 ± 0.0	15317.6 ± 0.0	9231.0 ± 8.4	10 [70]	10 [144]	10 [100]	7
Interp.⋆	56608.9 ± 2547.5	7484.9 ± 0.0	15317.6 ± 0.0	9231.0 ± 8.4	10 [70]	10 [144]	10 [100]	7
Linear Ramp⋆	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS	7459.6 ± 2169.4	6888.2 ± 0.0	14411.5 ± 0.0	9179.9 ± 9.1	10 [70]	10 [144]	10 [100]	7
TQA⋆	-	-	-	-	-	-	-	
TQA	-	-	-	-	-	-	-	
PP								
Fourier⋆	56682.2 ± 2535.8	7484.9 ± 0.0	15317.6 ± 0.0	9231.0 ± 8.4	10 [70]	10 [144]	10 [100]	7
Interp.⋆	56598.1 ± 2542.4	7484.9 ± 0.0	15317.6 ± 0.0	9231.0 ± 8.4	10 [70]	10 [144]	10 [100]	7
Linear Ramp⋆	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS	7459.6 ± 2169.4	6888.2 ± 0.0	14411.5 ± 0.0	9179.9 ± 9.1	10 [70]	10 [144]	10 [100]	7
TQA⋆	-	-	-	-	-	-	-	
TQA	-	-	-	-	-	-	-	
SV								
Fourier⋆	2839.4 ± 631.4	940.5 ± 0.0	7569.7 ± 4127.4	5998.2 ± 3052.2	10 [20]	10 [21]	10 [20]	7
Interp.⋆	2839.4 ± 631.4	940.5 ± 0.0	7569.7 ± 4127.3	5998.2 ± 3052.2	10 [20]	10 [21]	10 [20]	7
Linear Ramp⋆	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS	2839.4 ± 631.4	940.5 ± 0.0	7569.7 ± 4127.3	5998.2 ± 3052.2	10 [20]	10 [21]	10 [20]	7
TQA⋆	-	-	-	1874.4	-	-	-	1
TQA	-	-	-	1806.3	-	-	-	1

Table 33: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

12 Tables for depth $P = 2$

	ER	HH	LB	RR
	MPS (Aer)			
Fourier*	10 [70]	10 [144]	10 [100]	7 [100]
Interp.*	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	10 [70]	10 [144]	10 [100]	7 [100]
TQA*	10 [70]	10 [144]	10 [100]	7 [100]
TQA	10 [70]	10 [144]	10 [100]	7 [100]
	MPS (Quimb)			
Fourier*	10 [70]	10 [144]	10 [100]	7 [100]
Interp.*	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	10 [70]	10 [144]	10 [100]	7 [100]
TQA*	-	10 [144]	-	-
TQA	-	10 [144]	-	-
	PP			
Fourier*	10 [70]	10 [144]	10 [100]	7 [100]
Interp.*	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	10 [70]	10 [144]	10 [100]	7 [100]
TQA*	10 [70]	10 [144]	10 [100]	7 [100]
TQA	10 [70]	10 [144]	10 [100]	7 [100]
	SV			
Fourier*	10 [20]	10 [21]	10 [20]	7 [20]
Interp.*	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	10 [20]	10 [21]	10 [20]	7 [20]
TQA*	-	-	-	1 [20]
TQA	-	-	-	1 [20]

Table 34: **Number of Instances (num. nodes in brackets) for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fourier⋆	61936.6 ± 2524.1	15087.0 ± 19.3	23286.0 ± 577.7	16872.2 ± 120.9
Interp.⋆	53447.9 ± 6357.5	15047.8 ± 53.3	19285.1 ± 523.1	16980.0 ± 51.8
Linear Ramp⋆	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	56445.9 ± 3505.9	12041.7 ± 332.9	22230.0 ± 434.7	16865.3 ± 106.3
TQA⋆	59220.7 ± 3629.8	15043.9 ± 77.7	23709.2 ± 162.9	16999.8 ± 50.3
TQA	26859.1 ± 3717.5	11388.1 ± 8.8	21309.1 ± 17.0	15093.9 ± 51.3
MPS (Quimb)				
Fourier⋆	60904.9 ± 2594.3	15226.8 ± 37.0	23638.7 ± 1.6	16775.7 ± 129.0
Interp.⋆	57420.3 ± 3584.7	12861.5 ± 79.6	21874.6 ± 0.0	16888.8 ± 28.5
Linear Ramp⋆	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	57134.4 ± 2418.9	13032.0 ± 268.7	18115.6 ± 0.0	16677.2 ± 97.5
TQA⋆	-	15348.6 ± 272.1	-	-
TQA	-	13385.3 ± 0.0	-	-
PP				
Fourier⋆	66023.6 ± 2531.7	14667.9 ± 0.0	23962.1 ± 0.0	16648.0 ± 8.2
Interp.⋆	53045.1 ± 6171.8	14668.8 ± 0.0	20205.4 ± 0.0	16647.0 ± 8.6
Linear Ramp⋆	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	55072.4 ± 3444.0	10143.3 ± 0.0	17625.4 ± 0.0	16546.3 ± 7.9
TQA⋆	64848.8 ± 5913.4	14668.5 ± 0.0	23961.1 ± 0.0	16643.3 ± 9.1
TQA	23622.4 ± 2109.4	11591.9 ± 0.0	21688.0 ± 0.0	14559.1 ± 7.2
SV				
Fourier⋆	3945.4 ± 756.2	1853.2 ± 0.0	8305.3 ± 3734.0	7126.2 ± 2801.9
Interp.⋆	3859.0 ± 719.9	934.7 ± 0.0	7320.4 ± 2999.7	6426.5 ± 2497.8
Linear Ramp⋆	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	3922.0 ± 695.1	1379.1 ± 0.0	8429.1 ± 3760.8	6989.7 ± 2874.7
TQA⋆	-	-	-	3308.9
TQA	-	-	-	2918.4

Table 35: **Avg. Energy ± standard deviation for MIS for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy				Num. Instances		
	ER	HH	LB	RR	ER	HH	LB
MPS (Aer)							
Fourier⋆	61936.6 ± 2524.1	15087.0 ± 19.3	23286.0 ± 577.7	16872.2 ± 120.9	10 [70]	10 [144]	10 [100]
Interp.⋆	53447.9 ± 6357.5	15047.8 ± 53.3	19285.1 ± 523.1	16980.0 ± 51.8	10 [70]	10 [144]	10 [100]
Linear Ramp⋆	-	-	-	-	-	-	-
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp†	-	-	-	-	-	-	-
Recursive TS	56445.9 ± 3505.9	12041.7 ± 332.9	22230.0 ± 434.7	16865.3 ± 106.3	10 [70]	10 [144]	10 [100]
TQA⋆	59220.7 ± 3629.8	15043.9 ± 77.7	23709.2 ± 162.9	16999.8 ± 50.3	10 [70]	10 [144]	10 [100]
TQA	26859.1 ± 3717.5	11388.1 ± 8.8	21309.1 ± 17.0	15093.9 ± 51.3	10 [70]	10 [144]	10 [100]
MPS (Quimb)							
Fourier⋆	60904.9 ± 2594.3	15226.8 ± 37.0	23638.7 ± 1.6	16775.7 ± 129.0	10 [70]	10 [144]	10 [100]
Interp.⋆	57420.3 ± 3584.7	12861.5 ± 79.6	21874.6 ± 0.0	16888.8 ± 28.5	10 [70]	10 [144]	10 [100]
Linear Ramp⋆	-	-	-	-	-	-	-
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp†	-	-	-	-	-	-	-
Recursive TS	57134.4 ± 2418.9	13032.0 ± 268.7	18115.6 ± 0.0	16677.2 ± 97.5	10 [70]	10 [144]	10 [100]
TQA⋆	-	15348.6 ± 272.1	-	-	-	10 [144]	-
TQA	-	13385.3 ± 0.0	-	-	-	10 [144]	-
PP							
Fourier⋆	66023.6 ± 2531.7	14667.9 ± 0.0	23962.1 ± 0.0	16648.0 ± 8.2	10 [70]	10 [144]	10 [100]
Interp.⋆	53045.1 ± 6171.8	14668.8 ± 0.0	20205.4 ± 0.0	16647.0 ± 8.6	10 [70]	10 [144]	10 [100]
Linear Ramp⋆	-	-	-	-	-	-	-
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp†	-	-	-	-	-	-	-
Recursive TS	55072.4 ± 3444.0	10143.3 ± 0.0	17625.4 ± 0.0	16546.3 ± 7.9	10 [70]	10 [144]	10 [100]
TQA⋆	64848.8 ± 5913.4	14668.5 ± 0.0	23961.1 ± 0.0	16643.3 ± 9.1	10 [70]	10 [144]	10 [100]
TQA	23622.4 ± 2109.4	11591.9 ± 0.0	21688.0 ± 0.0	14559.1 ± 7.2	10 [70]	10 [144]	10 [100]
SV							
Fourier⋆	3945.4 ± 756.2	1853.2 ± 0.0	8305.3 ± 3734.0	7126.2 ± 2801.9	10 [20]	10 [21]	10 [20]
Interp.⋆	3859.0 ± 719.9	934.7 ± 0.0	7320.4 ± 2999.7	6426.5 ± 2497.8	10 [20]	10 [21]	10 [20]
Linear Ramp⋆	-	-	-	-	-	-	-
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp†	-	-	-	-	-	-	-
Recursive TS	3922.0 ± 695.1	1379.1 ± 0.0	8429.1 ± 3760.8	6989.7 ± 2874.7	10 [20]	10 [21]	10 [20]
TQA⋆	-	-	-	3308.9	-	-	-
TQA	-	-	-	2918.4	-	-	-

Table 36: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

13 Tables for depth $P = 3$

	ER	HH	LB	RR
		MPS (Aer)		
Fourier*	10 [70]	10 [144]	10 [100]	7 [100]
Interp.*	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	10 [70]	10 [144]	10 [100]	7 [100]
TQA*	-	-	-	-
TQA	-	-	-	-
		MPS (Quimb)		
Fourier*	10 [70]	10 [144]	10 [100]	7 [100]
Interp.*	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	10 [70]	10 [144]	10 [100]	7 [100]
TQA*	-	-	-	-
TQA	-	-	-	-
		PP		
Fourier*	10 [70]	10 [144]	10 [100]	7 [100]
Interp.*	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	10 [70]	10 [144]	10 [100]	7 [100]
TQA*	-	-	-	-
TQA	-	-	-	-
		SV		
Fourier*	10 [20]	10 [21]	10 [20]	7 [20]
Interp.*	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	10 [20]	10 [21]	10 [20]	7 [20]
TQA*	-	-	-	1 [20]
TQA	-	-	-	1 [20]

Table 37: **Number of Instances (num. nodes in brackets) for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
		MPS (Aer)		
Fourier*	63673.4 ± 2329.8	16820.8 ± 51.2	23195.5 ± 513.9	17131.9 ± 170.1
Interp.*	30291.4 ± 8828.4	15646.5 ± 437.4	20601.2 ± 388.8	17145.9 ± 139.2
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	62008.2 ± 2330.0	15070.2 ± 309.9	24138.8 ± 81.7	17167.3 ± 57.3
TQA*	-	-	-	-
TQA	-	-	-	-
		MPS (Quimb)		
Fourier*	62922.6 ± 2139.0	16451.2 ± 45.1	24273.5 ± 34.5	16612.2 ± 494.5
Interp.*	33146.6 ± 11917.7	12956.4 ± 124.4	17428.8 ± 0.0	17105.4 ± 115.2
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	61938.4 ± 2733.1	15063.6 ± 545.3	20583.1 ± 736.0	17031.2 ± 60.0
TQA*	-	-	-	-
TQA	-	-	-	-
		PP		
Fourier*	70899.1 ± 2921.4	16638.6 ± 0.0	25141.6 ± 0.0	17515.4 ± 3.4
Interp.*	15399.2 ± 5117.8	16615.9 ± 0.0	23006.9 ± 0.0	17489.3 ± 12.8
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	62957.6 ± 5057.1	15004.4 ± 0.0	24129.8 ± 0.0	17201.5 ± 120.1
TQA*	-	-	-	-
TQA	-	-	-	-
		SV		
Fourier*	4260.1 ± 742.6	2069.8 ± 0.0	8957.7 ± 4034.6	7576.3 ± 3083.8
Interp.*	3916.1 ± 687.1	1579.4 ± 0.0	5958.9 ± 3763.0	4536.5 ± 1679.9
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	4335.6 ± 684.4	1946.3 ± 0.0	8692.1 ± 3718.1	7331.9 ± 2682.5
TQA*	-	-	-	3483.0
TQA	-	-	-	3244.3

Table 38: **Avg. Energy ± standard deviation for MIS for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy				Num. Instances		
	ER	HH	LB	RR	ER	HH	LB
MPS (Aer)							
Fourier*	63673.4 ± 2329.8	16820.8 ± 51.2	23195.5 ± 513.9	17131.9 ± 170.1	10 [70]	10 [144]	10 [100]
Interp.*	30291.4 ± 8828.4	15646.5 ± 437.4	20601.2 ± 388.8	17145.9 ± 139.2	10 [70]	10 [144]	10 [100]
Linear Ramp*	-	-	-	-	-	-	-
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp†	-	-	-	-	-	-	-
Recursive TS	62008.2 ± 2330.0	15070.2 ± 309.9	24138.8 ± 81.7	17167.3 ± 57.3	10 [70]	10 [144]	10 [100]
TQA*	-	-	-	-	-	-	-
TQA	-	-	-	-	-	-	-
MPS (Quimb)							
Fourier*	62922.6 ± 2139.0	16451.2 ± 45.1	24273.5 ± 34.5	16612.2 ± 494.5	10 [70]	10 [144]	10 [100]
Interp.*	33146.6 ± 11917.7	12956.4 ± 124.4	17428.8 ± 0.0	17105.4 ± 115.2	10 [70]	10 [144]	10 [100]
Linear Ramp*	-	-	-	-	-	-	-
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp†	-	-	-	-	-	-	-
Recursive TS	61938.4 ± 2733.1	15063.6 ± 545.3	20583.1 ± 736.0	17031.2 ± 60.0	10 [70]	10 [144]	10 [100]
TQA*	-	-	-	-	-	-	-
TQA	-	-	-	-	-	-	-
PP							
Fourier*	70899.1 ± 2921.4	16638.6 ± 0.0	25141.6 ± 0.0	17515.4 ± 3.4	10 [70]	10 [144]	10 [100]
Interp.*	15399.2 ± 5117.8	16615.9 ± 0.0	23006.9 ± 0.0	17489.3 ± 12.8	10 [70]	10 [144]	10 [100]
Linear Ramp*	-	-	-	-	-	-	-
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp†	-	-	-	-	-	-	-
Recursive TS	62957.6 ± 5057.1	15004.4 ± 0.0	24129.8 ± 0.0	17201.5 ± 120.1	10 [70]	10 [144]	10 [100]
TQA*	-	-	-	-	-	-	-
TQA	-	-	-	-	-	-	-
SV							
Fourier*	4260.1 ± 742.6	2069.8 ± 0.0	8957.7 ± 4034.6	7576.3 ± 3083.8	10 [20]	10 [21]	10 [20]
Interp.*	3916.1 ± 687.1	1579.4 ± 0.0	5958.9 ± 3763.0	4536.5 ± 1679.9	10 [20]	10 [21]	10 [20]
Linear Ramp*	-	-	-	-	-	-	-
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp†	-	-	-	-	-	-	-
Recursive TS	4335.6 ± 684.4	1946.3 ± 0.0	8692.1 ± 3718.1	7331.9 ± 2682.5	10 [20]	10 [21]	10 [20]
TQA*	-	-	-	3483.0	-	-	-
TQA	-	-	-	3244.3	-	-	-

Table 39: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

14 Tables for depth $P = 4$

	ER	HH	LB	RR
	MPS (Aer)			
Fourier*	10 [70]	10 [144]	10 [100]	7 [100]
Interp.*	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	10 [70]	10 [144]	10 [100]	7 [100]
TQA*	10 [70]	10 [144]	10 [100]	7 [100]
TQA	10 [70]	10 [144]	10 [100]	7 [100]
	MPS (Quimb)			
Fourier*	10 [70]	10 [144]	10 [100]	7 [100]
Interp.*	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	10 [70]	10 [144]	10 [100]	7 [100]
TQA*	-	10 [144]	-	-
TQA	-	10 [144]	-	-
	PP			
Fourier*	10 [70]	10 [144]	10 [100]	7 [100]
Interp.*	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	10 [70]	10 [144]	10 [100]	7 [100]
TQA*	10 [70]	10 [144]	10 [100]	7 [100]
TQA	10 [70]	10 [144]	10 [100]	7 [100]
	SV			
Fourier*	10 [20]	10 [21]	10 [20]	7 [20]
Interp.*	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	10 [20]	10 [21]	10 [20]	7 [20]
TQA*	-	-	-	1 [20]
TQA	-	-	-	1 [20]

Table 40: **Number of Instances (num. nodes in brackets) for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
		MPS (Aer)		
Fourier*	65073.4 ± 2065.2	17025.1 ± 65.9	23889.9 ± 331.7	17236.2 ± 127.9
Interp.*	30289.6 ± 18260.2	16319.8 ± 349.4	22168.3 ± 982.7	17346.4 ± 56.5
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	63848.3 ± 2289.2	15798.4 ± 470.4	24240.1 ± 58.4	17209.0 ± 47.0
TQA*	52894.8 ± 6299.2	16825.6 ± 75.2	23803.7 ± 314.5	17315.5 ± 23.9
TQA	45337.4 ± 4065.1	15942.9 ± 6.2	21463.2 ± 18.7	16589.3 ± 17.4
		MPS (Quimb)		
Fourier*	64589.8 ± 2235.3	16692.3 ± 121.3	24594.6 ± 16.6	16867.9 ± 357.5
Interp.*	19813.9 ± 15691.9	14400.9 ± 631.7	19699.2 ± 0.0	17252.0 ± 58.5
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	63380.3 ± 2539.4	16097.6 ± 213.6	22181.1 ± 668.0	17178.1 ± 25.5
TQA*	-	16122.0 ± 113.1	-	-
TQA	-	15660.4 ± 8.7	-	-
		PP		
Fourier*	73002.8 ± 3089.2	17057.3 ± 0.0	25757.2 ± 0.0	17619.8 ± 16.1
Interp.*	27879.7 ± 27264.2	16074.5 ± 0.0	23790.6 ± 0.0	17581.2 ± 17.6
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	70768.3 ± 9211.8	15437.9 ± 0.0	24339.6 ± 0.0	17510.0 ± 5.1
TQA*	57385.8 ± 12377.5	16902.2 ± 0.0	24749.9 ± 0.0	17513.7 ± 22.3
TQA	18316.7 ± 5870.9	16094.1 ± 0.0	21509.8 ± 0.0	16272.7 ± 3.4
		SV		
Fourier*	4517.7 ± 742.2	2204.0 ± 0.0	9144.0 ± 4065.7	7776.5 ± 3129.3
Interp.*	4042.6 ± 642.8	1280.9 ± 0.0	5135.8 ± 3812.9	4016.3 ± 2128.0
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	4469.5 ± 698.7	1982.8 ± 0.0	8986.2 ± 3812.9	7659.6 ± 3026.5
TQA*	-	-	-	3479.1
TQA	-	-	-	3259.2

Table 41: **Avg. Energy ± standard deviation for MIS for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy				Num. Instances		
	ER	HH	LB	RR	ER	HH	LB
			MPS (Aer)				
Fourier*	65073.4 ± 2065.2	17025.1 ± 65.9	23889.9 ± 331.7	17236.2 ± 127.9	10 [70]	10 [144]	10 [100]
Interp.*	30289.6 ± 18260.2	16319.8 ± 349.4	22168.3 ± 982.7	17346.4 ± 56.5	10 [70]	10 [144]	10 [100]
Linear Ramp*	-	-	-	-	-	-	-
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp†	-	-	-	-	-	-	-
Recursive TS	63848.3 ± 2289.2	15798.4 ± 470.4	24240.1 ± 58.4	17209.0 ± 47.0	10 [70]	10 [144]	10 [100]
TQA*	52894.8 ± 6299.2	16825.6 ± 75.2	23803.7 ± 314.5	17315.5 ± 23.9	10 [70]	10 [144]	10 [100]
TQA	45337.4 ± 4065.1	15942.9 ± 6.2	21463.2 ± 18.7	16589.3 ± 17.4	10 [70]	10 [144]	10 [100]
			MPS (Quimb)				
Fourier*	64589.8 ± 2235.3	16692.3 ± 121.3	24594.6 ± 16.6	16867.9 ± 357.5	10 [70]	10 [144]	10 [100]
Interp.*	19813.9 ± 15691.9	14400.9 ± 631.7	19699.2 ± 0.0	17252.0 ± 58.5	10 [70]	10 [144]	10 [100]
Linear Ramp*	-	-	-	-	-	-	-
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp†	-	-	-	-	-	-	-
Recursive TS	63380.3 ± 2539.4	16097.6 ± 213.6	22181.1 ± 668.0	17178.1 ± 25.5	10 [70]	10 [144]	10 [100]
TQA*	-	16122.0 ± 113.1	-	-	-	10 [144]	-
TQA	-	15660.4 ± 8.7	-	-	-	10 [144]	-
			PP				
Fourier*	73002.8 ± 3089.2	17057.3 ± 0.0	25757.2 ± 0.0	17619.8 ± 16.1	10 [70]	10 [144]	10 [100]
Interp.*	27879.7 ± 27264.2	16074.5 ± 0.0	23790.6 ± 0.0	17581.2 ± 17.6	10 [70]	10 [144]	10 [100]
Linear Ramp*	-	-	-	-	-	-	-
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp†	-	-	-	-	-	-	-
Recursive TS	70768.3 ± 9211.8	15437.9 ± 0.0	24339.6 ± 0.0	17510.0 ± 5.1	10 [70]	10 [144]	10 [100]
TQA*	57385.8 ± 12377.5	16902.2 ± 0.0	24749.9 ± 0.0	17513.7 ± 22.3	10 [70]	10 [144]	10 [100]
TQA	18316.7 ± 5870.9	16094.1 ± 0.0	21509.8 ± 0.0	16272.7 ± 3.4	10 [70]	10 [144]	10 [100]
			SV				
Fourier*	4517.7 ± 742.2	2204.0 ± 0.0	9144.0 ± 4065.7	7776.5 ± 3129.3	10 [20]	10 [21]	10 [20]
Interp.*	4042.6 ± 642.8	1280.9 ± 0.0	5135.8 ± 3812.9	4016.3 ± 2128.0	10 [20]	10 [21]	10 [20]
Linear Ramp*	-	-	-	-	-	-	-
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp†	-	-	-	-	-	-	-
Recursive TS	4469.5 ± 698.7	1982.8 ± 0.0	8986.2 ± 3812.9	7659.6 ± 3026.5	10 [20]	10 [21]	10 [20]
TQA*	-	-	-	3479.1	-	-	-
TQA	-	-	-	3259.2	-	-	-

Table 42: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

15 Tables for depth $P = 5$

	ER	HH	LB	RR
		MPS (Aer)		
Fourier*	10 [70]	10 [144]	10 [100]	7 [100]
Interp.*	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	10 [70]	10 [144]	10 [100]	7 [100]
TQA*	-	-	-	-
TQA	-	-	-	-
		MPS (Quimb)		
Fourier*	10 [70]	10 [144]	10 [100]	7 [100]
Interp.*	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	10 [70]	10 [144]	10 [100]	7 [100]
TQA*	-	-	-	-
TQA	-	-	-	-
		PP		
Fourier*	10 [70]	10 [144]	10 [100]	7 [100]
Interp.*	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	10 [70]	10 [144]	10 [100]	7 [100]
TQA*	-	-	-	-
TQA	-	-	-	-
		SV		
Fourier*	10 [20]	10 [21]	10 [20]	7 [20]
Interp.*	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	10 [20]	10 [21]	10 [20]	7 [20]
TQA*	-	-	-	1 [20]
TQA	-	-	-	1 [20]

Table 43: **Number of Instances (num. nodes in brackets) for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fourier*	65502.3 ± 2332.4	17150.3 ± 42.1	24101.5 ± 259.9	17354.7 ± 83.4
Interp.*	27765.9 ± 13747.4	16526.5 ± 257.7	22601.5 ± 712.6	17426.1 ± 46.1
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	64889.2 ± 2226.7	16096.0 ± 514.5	24296.4 ± 39.2	17239.8 ± 47.2
TQA*	-	-	-	-
TQA	-	-	-	-
MPS (Quimb)				
Fourier*	64437.4 ± 4439.6	17080.5 ± 86.0	24634.6 ± 6.7	17070.4 ± 306.2
Interp.*	13283.3 ± 19301.9	14645.9 ± 444.9	21688.4 ± 0.0	17255.5 ± 120.8
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	64315.6 ± 2555.1	16265.4 ± 232.2	23287.8 ± 384.9	17212.6 ± 25.3
TQA*	-	-	-	-
TQA	-	-	-	-
PP				
Fourier*	79758.6 ± 2424.4	17143.2 ± 0.0	26076.0 ± 0.0	17773.5 ± 18.9
Interp.*	35592.9 ± 25747.1	16105.4 ± 0.0	19595.6 ± 0.0	17759.5 ± 29.4
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	79043.0 ± 14470.5	16317.0 ± 0.0	24697.4 ± 0.0	17589.3 ± 28.1
TQA*	-	-	-	-
TQA	-	-	-	-
SV				
Fourier*	4652.9 ± 746.8	2217.9 ± 0.0	9295.9 ± 4091.1	7870.5 ± 3186.7
Interp.*	4078.6 ± 624.5	1440.4 ± 0.0	4298.4 ± 2632.0	4072.0 ± 2156.8
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	4533.5 ± 714.4	1991.6 ± 0.0	9066.1 ± 3849.1	7772.1 ± 3092.2
TQA*	-	-	-	3476.4
TQA	-	-	-	3308.0

Table 44: **Avg. Energy $\pm$ standard deviation for MIS for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy				Num. Instances		
	ER	HH	LB	RR	ER	HH	LB
				MPS (Aer)			
Fourier*	65502.3 ± 2332.4	17150.3 ± 42.1	24101.5 ± 259.9	17354.7 ± 83.4	10 [70]	10 [144]	10 [100]
Interp.*	27765.9 ± 13747.4	16526.5 ± 257.7	22601.5 ± 712.6	17426.1 ± 46.1	10 [70]	10 [144]	10 [100]
Linear Ramp*	-	-	-	-	-	-	-
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp†	-	-	-	-	-	-	-
Recursive TS	64889.2 ± 2226.7	16096.0 ± 514.5	24296.4 ± 39.2	17239.8 ± 47.2	10 [70]	10 [144]	10 [100]
TQA*	-	-	-	-	-	-	-
TQA	-	-	-	-	-	-	-
				MPS (Quimb)			
Fourier*	64437.4 ± 4439.6	17080.5 ± 86.0	24634.6 ± 6.7	17070.4 ± 306.2	10 [70]	10 [144]	10 [100]
Interp.*	13283.3 ± 19301.9	14645.9 ± 444.9	21688.4 ± 0.0	17255.5 ± 120.8	10 [70]	10 [144]	10 [100]
Linear Ramp*	-	-	-	-	-	-	-
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp†	-	-	-	-	-	-	-
Recursive TS	64315.6 ± 2555.1	16265.4 ± 232.2	23287.8 ± 384.9	17212.6 ± 25.3	10 [70]	10 [144]	10 [100]
TQA*	-	-	-	-	-	-	-
TQA	-	-	-	-	-	-	-
				PP			
Fourier*	79758.6 ± 2424.4	17143.2 ± 0.0	26076.0 ± 0.0	17773.5 ± 18.9	10 [70]	10 [144]	10 [100]
Interp.*	35592.9 ± 25747.1	16105.4 ± 0.0	19595.6 ± 0.0	17759.5 ± 29.4	10 [70]	10 [144]	10 [100]
Linear Ramp*	-	-	-	-	-	-	-
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp†	-	-	-	-	-	-	-
Recursive TS	79043.0 ± 14470.5	16317.0 ± 0.0	24697.4 ± 0.0	17589.3 ± 28.1	10 [70]	10 [144]	10 [100]
TQA*	-	-	-	-	-	-	-
TQA	-	-	-	-	-	-	-
				SV			
Fourier*	4652.9 ± 746.8	2217.9 ± 0.0	9295.9 ± 4091.1	7870.5 ± 3186.7	10 [20]	10 [21]	10 [20]
Interp.*	4078.6 ± 624.5	1440.4 ± 0.0	4298.4 ± 2632.0	4072.0 ± 2156.8	10 [20]	10 [21]	10 [20]
Linear Ramp*	-	-	-	-	-	-	-
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp†	-	-	-	-	-	-	-
Recursive TS	4533.5 ± 714.4	1991.6 ± 0.0	9066.1 ± 3849.1	7772.1 ± 3092.2	10 [20]	10 [21]	10 [20]
TQA*	-	-	-	3476.4	-	-	-
TQA	-	-	-	3308.0	-	-	-

Table 45: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

16 Tables for depth $P = 6$

	ER	HH	LB	RR
		MPS (Aer)		
Fourier⋆	10 [70]	10 [144]	10 [100]	7 [100]
Interp.⋆	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp⋆	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [70]	10 [144]	10 [100]	7 [100]
Recursive TS	10 [70]	10 [144]	10 [100]	7 [100]
TQA⋆	10 [70]	10 [144]	10 [100]	7 [100]
TQA	10 [70]	10 [144]	10 [100]	7 [100]
		MPS (Quimb)		
Fourier⋆	10 [70]	10 [144]	10 [100]	7 [100]
Interp.⋆	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp⋆	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [70]	10 [144]	10 [100]	7 [100]
Recursive TS	10 [70]	10 [144]	10 [100]	7 [100]
TQA⋆	10 [70]	10 [144]	10 [100]	7 [100]
TQA	10 [70]	10 [144]	10 [100]	7 [100]
		PP		
Fourier⋆	10 [70]	10 [144]	10 [100]	7 [100]
Interp.⋆	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp⋆	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [70]	10 [144]	10 [100]	7 [100]
Recursive TS	10 [70]	10 [144]	10 [100]	7 [100]
TQA⋆	10 [70]	10 [144]	10 [100]	7 [100]
TQA	10 [70]	10 [144]	10 [100]	7 [100]
		SV		
Fourier⋆	10 [20]	10 [21]	10 [20]	7 [20]
Interp.⋆	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp⋆	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp†	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS	10 [20]	10 [21]	10 [20]	7 [20]
TQA⋆	10 [20]	10 [21]	10 [20]	7 [20]
TQA	10 [20]	10 [21]	10 [20]	7 [20]

Table 46: **Number of Instances (num. nodes in brackets) for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fourier*	65882.6 ± 2605.2	17238.8 ± 41.9	24117.0 ± 532.0	17396.2 ± 68.4
Interp.*	41477.0 ± 10764.6	16407.2 ± 251.6	22849.9 ± 266.4	17439.4 ± 59.8
Linear Ramp*	63828.8 ± 1920.9	17199.0 ± 19.1	24453.7 ± 62.5	17474.2 ± 19.2
Linear Ramp	58099.4 ± 5174.1	16689.6 ± 9.6	24088.6 ± 117.4	16943.9 ± 14.9
Linear Ramp†	13260.6 ± 3107.3	15020.7 ± 16.9	19940.6 ± 24.3	15035.5 ± 18.1
Recursive TS	65385.0 ± 2303.4	16192.4 ± 543.3	24319.1 ± 40.1	17263.4 ± 56.7
TQA*	32061.9 ± 11764.8	17136.6 ± 34.3	24234.3 ± 142.4	17302.5 ± 39.1
TQA	12617.7 ± 2917.7	16331.0 ± 4.1	21160.4 ± 21.2	16776.6 ± 9.0
MPS (Quimb)				
Fourier*	65029.4 ± 4379.9	17218.3 ± 28.1	24645.5 ± 21.4	17161.4 ± 323.8
Interp.*	11970.4 ± 10406.1	14789.6 ± 641.0	21925.2 ± 82.9	17251.8 ± 122.1
Linear Ramp*	61009.2 ± 10563.0	17138.4 ± 42.8	24320.1 ± 45.3	17464.3 ± 10.6
Linear Ramp	54526.2 ± 16538.3	16413.5 ± 0.2	23860.1 ± 2.0	16919.7 ± 7.4
Linear Ramp†	12773.3 ± 5743.0	14894.2 ± 0.0	18578.1 ± 0.0	15067.9 ± 13.3
Recursive TS	64715.2 ± 2579.0	16341.1 ± 239.8	23642.7 ± 120.3	17237.7 ± 19.7
TQA*	44900.5 ± 13578.2	17190.4 ± 15.8	23112.6 ± 0.4	17295.8 ± 27.9
TQA	21946.0 ± 12600.0	16300.9 ± 0.7	18395.7 ± 0.0	16773.6 ± 16.5
PP				
Fourier*	78103.2 ± 7976.1	17172.0 ± 0.0	25853.4 ± 0.0	17937.1 ± 32.6
Interp.*	55315.5 ± 32067.7	16369.2 ± 0.0	25263.7 ± 0.0	17782.2 ± 19.1
Linear Ramp*	52197.1 ± 5099.0	17171.5 ± 0.0	25640.2 ± 0.0	17780.5 ± 36.5
Linear Ramp	25132.2 ± 3849.8	16555.8 ± 0.0	24967.6 ± 0.0	17039.4 ± 14.4
Linear Ramp†	-523.6 ± 6928.0	14551.0 ± 0.0	18733.9 ± 0.0	14501.0 ± 1.5
Recursive TS	81317.8 ± 13668.3	16529.8 ± 0.0	24697.8 ± 0.0	17612.9 ± 40.0
TQA*	19827.1 ± 3918.3	17075.8 ± 0.0	25357.7 ± 0.0	17632.6 ± 96.0
TQA	3285.8 ± 753.9	15990.3 ± 0.0	18897.2 ± 0.0	16842.6 ± 4.4
SV				
Fourier*	4716.6 ± 742.4	2212.8 ± 0.0	9290.9 ± 4162.0	7664.8 ± 3014.5
Interp.*	4089.5 ± 607.9	1711.2 ± 0.0	4905.3 ± 3834.9	3880.6 ± 2462.0
Linear Ramp*	4829.3 ± 792.9	2231.4 ± 0.0	9282.9 ± 3910.1	7918.1 ± 3162.0
Linear Ramp	4777.5 ± 779.0	2107.4 ± 0.0	6402.1 ± 2074.9	6448.6 ± 2419.1
Linear Ramp†	3542.3 ± 359.6	1847.2 ± 0.0	4839.5 ± 1178.5	4470.3 ± 970.5
Recursive TS	4578.2 ± 720.8	2094.1 ± 0.0	9099.4 ± 3870.4	7803.1 ± 3117.3
TQA*	4239.2 ± 561.7	2228.6 ± 0.0	6051.2 ± 1720.6	6584.1 ± 2329.4
TQA	2644.2 ± 630.6	2126.2 ± 0.0	2784.5 ± 831.0	4028.6 ± 1065.1

Table 47: **Avg. Energy $\pm$ standard deviation for MIS for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy				Num. Instances		
	ER	HH	LB	RR	ER	HH	LB
MPS (Aer)							
Fourier*	65882.6 ± 2605.2	17238.8 ± 41.9	24117.0 ± 532.0	17396.2 ± 68.4	10 [70]	10 [144]	10 [100]
Interp.*	41477.0 ± 10764.6	16407.2 ± 251.6	22849.9 ± 266.4	17439.4 ± 59.8	10 [70]	10 [144]	10 [100]
Linear Ramp*	63828.8 ± 1920.9	17199.0 ± 19.1	24453.7 ± 62.5	17474.2 ± 19.2	10 [70]	10 [144]	10 [100]
Linear Ramp	58099.4 ± 5174.1	16689.6 ± 9.6	24088.6 ± 117.4	16943.9 ± 14.9	10 [70]	10 [144]	10 [100]
Linear Ramp†	13260.6 ± 3107.3	15020.7 ± 16.9	19940.6 ± 24.3	15035.5 ± 18.1	10 [70]	10 [144]	10 [100]
Recursive TS	65385.0 ± 2303.4	16192.4 ± 543.3	24319.1 ± 40.1	17263.4 ± 56.7	10 [70]	10 [144]	10 [100]
TQA*	32061.9 ± 11764.8	17136.6 ± 34.3	24234.3 ± 142.4	17302.5 ± 39.1	10 [70]	10 [144]	10 [100]
TQA	12617.7 ± 2917.7	16331.0 ± 4.1	21160.4 ± 21.2	16776.6 ± 9.0	10 [70]	10 [144]	10 [100]
MPS (Quimb)							
Fourier*	65029.4 ± 4379.9	17218.3 ± 28.1	24645.5 ± 21.4	17161.4 ± 323.8	10 [70]	10 [144]	10 [100]
Interp.*	11970.4 ± 10406.1	14789.6 ± 641.0	21925.2 ± 82.9	17251.8 ± 122.1	10 [70]	10 [144]	10 [100]
Linear Ramp*	61009.2 ± 10563.0	17138.4 ± 42.8	24320.1 ± 45.3	17464.3 ± 10.6	10 [70]	10 [144]	10 [100]
Linear Ramp	54526.2 ± 16538.3	16413.5 ± 0.2	23860.1 ± 2.0	16919.7 ± 7.4	10 [70]	10 [144]	10 [100]
Linear Ramp†	12773.3 ± 5743.0	14894.2 ± 0.0	18578.1 ± 0.0	15067.9 ± 13.3	10 [70]	10 [144]	10 [100]
Recursive TS	64715.2 ± 2579.0	16341.1 ± 239.8	23642.7 ± 120.3	17237.7 ± 19.7	10 [70]	10 [144]	10 [100]
TQA*	44900.5 ± 13578.2	17190.4 ± 15.8	23112.6 ± 0.4	17295.8 ± 27.9	10 [70]	10 [144]	10 [100]
TQA	21946.0 ± 12600.0	16300.9 ± 0.7	18395.7 ± 0.0	16773.6 ± 16.5	10 [70]	10 [144]	10 [100]
PP							
Fourier*	78103.2 ± 7976.1	17172.0 ± 0.0	25853.4 ± 0.0	17937.1 ± 32.6	10 [70]	10 [144]	10 [100]
Interp.*	55315.5 ± 32067.7	16369.2 ± 0.0	25263.7 ± 0.0	17782.2 ± 19.1	10 [70]	10 [144]	10 [100]
Linear Ramp*	52197.1 ± 5099.0	17171.5 ± 0.0	25640.2 ± 0.0	17780.5 ± 36.5	10 [70]	10 [144]	10 [100]
Linear Ramp	25132.2 ± 3849.8	16555.8 ± 0.0	24967.6 ± 0.0	17039.4 ± 14.4	10 [70]	10 [144]	10 [100]
Linear Ramp†	-523.6 ± 6928.0	14551.0 ± 0.0	18733.9 ± 0.0	14501.0 ± 1.5	10 [70]	10 [144]	10 [100]
Recursive TS	81317.8 ± 13668.3	16529.8 ± 0.0	24697.8 ± 0.0	17612.9 ± 40.0	10 [70]	10 [144]	10 [100]
TQA*	19827.1 ± 3918.3	17075.8 ± 0.0	25357.7 ± 0.0	17632.6 ± 96.0	10 [70]	10 [144]	10 [100]
TQA	3285.8 ± 753.9	15990.3 ± 0.0	18897.2 ± 0.0	16842.6 ± 4.4	10 [70]	10 [144]	10 [100]
SV							
Fourier*	4716.6 ± 742.4	2212.8 ± 0.0	9290.9 ± 4162.0	7664.8 ± 3014.5	10 [20]	10 [21]	10 [20]
Interp.*	4089.5 ± 607.9	1711.2 ± 0.0	4905.3 ± 3834.9	3880.6 ± 2462.0	10 [20]	10 [21]	10 [20]
Linear Ramp*	4829.3 ± 792.9	2231.4 ± 0.0	9282.9 ± 3910.1	7918.1 ± 3162.0	10 [20]	10 [21]	10 [20]
Linear Ramp	4777.5 ± 779.0	2107.4 ± 0.0	6402.1 ± 2074.9	6448.6 ± 2419.1	10 [20]	10 [21]	10 [20]
Linear Ramp†	3542.3 ± 359.6	1847.2 ± 0.0	4839.5 ± 1178.5	4470.3 ± 970.5	10 [20]	10 [21]	10 [20]
Recursive TS	4578.2 ± 720.8	2094.1 ± 0.0	9099.4 ± 3870.4	7803.1 ± 3117.3	10 [20]	10 [21]	10 [20]
TQA*	4239.2 ± 561.7	2228.6 ± 0.0	6051.2 ± 1720.6	6584.1 ± 2329.4	10 [20]	10 [21]	10 [20]
TQA	2644.2 ± 630.6	2126.2 ± 0.0	2784.5 ± 831.0	4028.6 ± 1065.1	10 [20]	10 [21]	10 [20]

Table 48: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

17 Tables for depth $P = 7$

	ER	HH	LB	RR
MPS (Aer)				
Fourier*	-	10 [144]	-	-
Interp.*	-	10 [144]	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	-	-	-	-
TQA*	-	-	-	-
TQA	-	-	-	-
MPS (Quimb)				
Fourier*	-	-	-	-
Interp.*	-	-	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	-	-	-	-
TQA*	-	-	-	-
TQA	-	-	-	-
PP				
Fourier*	-	10 [144]	-	-
Interp.*	-	10 [144]	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	-	-	-	-
TQA*	-	-	-	-
TQA	-	-	-	-
SV				
Fourier*	10 [20]	10 [21]	10 [20]	7 [20]
Interp.*	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	10 [20]	10 [21]	10 [20]	7 [20]
TQA*	-	-	-	1 [20]
TQA	-	-	-	1 [20]

Table 49: **Number of Instances (num. nodes in brackets) for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fourier⋆	-	17289.5 ± 51.9	-	-
Interp.⋆	-	16398.9 ± 190.0	-	-
Linear Ramp⋆	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	-	-	-	-
TQA⋆	-	-	-	-
TQA	-	-	-	-
MPS (Quimb)				
Fourier⋆	-	-	-	-
Interp.⋆	-	-	-	-
Linear Ramp⋆	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	-	-	-	-
TQA⋆	-	-	-	-
TQA	-	-	-	-
PP				
Fourier⋆	-	17176.5 ± 0.0	-	-
Interp.⋆	-	16528.1 ± 0.0	-	-
Linear Ramp⋆	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	-	-	-	-
TQA⋆	-	-	-	-
TQA	-	-	-	-
SV				
Fourier⋆	4747.8 ± 745.9	2208.8 ± 0.0	9028.3 ± 3854.4	7571.2 ± 2958.1
Interp.⋆	4100.8 ± 606.4	1640.5 ± 0.0	4115.2 ± 3116.2	3306.0 ± 2667.3
Linear Ramp⋆	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	4605.1 ± 707.9	2114.6 ± 0.0	9138.6 ± 3920.3	7845.0 ± 3152.0
TQA⋆	-	-	-	3490.7
TQA	-	-	-	3258.0

Table 50: **Avg. Energy $\pm$ standard deviation for MIS for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	R
MPS (Aer)								
Fourier⋆	-	17289.5 ± 51.9	-	-	-	10 [144]	-	
Interp.⋆	-	16398.9 ± 190.0	-	-	-	10 [144]	-	
Linear Ramp⋆	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS	-	-	-	-	-	-	-	
TQA⋆	-	-	-	-	-	-	-	
TQA	-	-	-	-	-	-	-	
MPS (Quimb)								
Fourier⋆	-	-	-	-	-	-	-	
Interp.⋆	-	-	-	-	-	-	-	
Linear Ramp⋆	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS	-	-	-	-	-	-	-	
TQA⋆	-	-	-	-	-	-	-	
TQA	-	-	-	-	-	-	-	
PP								
Fourier⋆	-	17176.5 ± 0.0	-	-	-	10 [144]	-	
Interp.⋆	-	16528.1 ± 0.0	-	-	-	10 [144]	-	
Linear Ramp⋆	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS	-	-	-	-	-	-	-	
TQA⋆	-	-	-	-	-	-	-	
TQA	-	-	-	-	-	-	-	
SV								
Fourier⋆	4747.8 ± 745.9	2208.8 ± 0.0	9028.3 ± 3854.4	7571.2 ± 2958.1	10 [20]	10 [21]	10 [20]	7
Interp.⋆	4100.8 ± 606.4	1640.5 ± 0.0	4115.2 ± 3116.2	3306.0 ± 2667.3	10 [20]	10 [21]	10 [20]	7
Linear Ramp⋆	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS	4605.1 ± 707.9	2114.6 ± 0.0	9138.6 ± 3920.3	7845.0 ± 3152.0	10 [20]	10 [21]	10 [20]	7
TQA⋆	-	-	-	3490.7	-	-	-	1
TQA	-	-	-	3258.0	-	-	-	1

Table 51: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

18 Tables for depth $P = 8$

	ER	HH	LB	RR
		MPS (Aer)		
Fourier*	-	10 [144]	-	-
Interp.*	-	10 [144]	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	-	-	-	-
TQA*	-	10 [144]	-	-
TQA	-	10 [144]	-	-
		MPS (Quimb)		
Fourier*	-	-	-	-
Interp.*	-	-	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	-	-	-	-
TQA*	-	10 [144]	-	-
TQA	-	10 [144]	-	-
		PP		
Fourier*	-	10 [144]	-	-
Interp.*	-	10 [144]	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	-	-	-	-
TQA*	-	10 [144]	-	-
TQA	-	10 [144]	-	-
		SV		
Fourier*	10 [20]	10 [21]	10 [20]	7 [20]
Interp.*	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	10 [20]	10 [21]	10 [20]	7 [20]
TQA*	-	-	-	1 [20]
TQA	-	-	-	1 [20]

Table 52: **Number of Instances (num. nodes in brackets) for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
		MPS (Aer)		
Fourier⋆	-	17315.2 ± 34.1	-	-
Interp.⋆	-	16279.0 ± 420.1	-	-
Linear Ramp⋆	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	-	-	-	-
TQA⋆	-	17252.6 ± 20.1	-	-
TQA	-	17132.3 ± 3.1	-	-
		MPS (Quimb)		
Fourier⋆	-	-	-	-
Interp.⋆	-	-	-	-
Linear Ramp⋆	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	-	-	-	-
TQA⋆	-	17070.9 ± 43.3	-	-
TQA	-	16609.5 ± 2.2	-	-
		PP		
Fourier⋆	-	17161.2 ± 0.0	-	-
Interp.⋆	-	16582.6 ± 0.0	-	-
Linear Ramp⋆	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	-	-	-	-
TQA⋆	-	17137.5 ± 0.1	-	-
TQA	-	16788.3 ± 0.0	-	-
		SV		
Fourier⋆	4758.6 ± 746.4	2168.6 ± 0.0	9217.4 ± 3993.7	7590.1 ± 3001.8
Interp.⋆	4094.5 ± 620.2	1600.4 ± 0.0	4043.8 ± 3198.8	3025.0 ± 2890.9
Linear Ramp⋆	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	4616.3 ± 710.9	2120.1 ± 0.0	9206.3 ± 4027.3	7866.3 ± 3170.9
TQA⋆	-	-	-	3451.4
TQA	-	-	-	3327.0

Table 53: **Avg. Energy $\pm$ standard deviation for MIS for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	R
MPS (Aer)								
Fourier*	-	17315.2 ± 34.1	-	-	-	10 [144]	-	
Interp.*	-	16279.0 ± 420.1	-	-	-	10 [144]	-	
Linear Ramp*	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS	-	-	-	-	-	-	-	
TQA*	-	17252.6 ± 20.1	-	-	-	10 [144]	-	
TQA	-	17132.3 ± 3.1	-	-	-	10 [144]	-	
MPS (Quimb)								
Fourier*	-	-	-	-	-	-	-	
Interp.*	-	-	-	-	-	-	-	
Linear Ramp*	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS	-	-	-	-	-	-	-	
TQA*	-	17070.9 ± 43.3	-	-	-	10 [144]	-	
TQA	-	16609.5 ± 2.2	-	-	-	10 [144]	-	
PP								
Fourier*	-	17161.2 ± 0.0	-	-	-	10 [144]	-	
Interp.*	-	16582.6 ± 0.0	-	-	-	10 [144]	-	
Linear Ramp*	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS	-	-	-	-	-	-	-	
TQA*	-	17137.5 ± 0.1	-	-	-	10 [144]	-	
TQA	-	16788.3 ± 0.0	-	-	-	10 [144]	-	
SV								
Fourier*	4758.6 ± 746.4	2168.6 ± 0.0	9217.4 ± 3993.7	7590.1 ± 3001.8	10 [20]	10 [21]	10 [20]	7
Interp.*	4094.5 ± 620.2	1600.4 ± 0.0	4043.8 ± 3198.8	3025.0 ± 2890.9	10 [20]	10 [21]	10 [20]	7
Linear Ramp*	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS	4616.3 ± 710.9	2120.1 ± 0.0	9206.3 ± 4027.3	7866.3 ± 3170.9	10 [20]	10 [21]	10 [20]	7
TQA*	-	-	-	3451.4	-	-	-	1
TQA	-	-	-	3327.0	-	-	-	1

Table 54: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

19 Tables for depth $P = 9$

	ER	HH	LB	RR
MPS (Aer)				
Fourier*	-	10 [144]	-	-
Interp.*	-	10 [144]	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	-	-	-	-
TQA*	-	-	-	-
TQA	-	-	-	-
MPS (Quimb)				
Fourier*	-	-	-	-
Interp.*	-	-	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	-	-	-	-
TQA*	-	-	-	-
TQA	-	-	-	-
PP				
Fourier*	-	10 [144]	-	-
Interp.*	-	10 [144]	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	-	-	-	-
TQA*	-	-	-	-
TQA	-	-	-	-
SV				
Fourier*	10 [20]	10 [21]	10 [20]	7 [20]
Interp.*	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	10 [20]	10 [21]	10 [20]	7 [20]
TQA*	-	-	-	1 [20]
TQA	-	-	-	1 [20]

Table 55: **Number of Instances (num. nodes in brackets) for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
		MPS (Aer)		
Fourier*	-	17318.8 ± 27.4	-	-
Interp.*	-	16162.6 ± 379.1	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	-	-	-	-
TQA*	-	-	-	-
TQA	-	-	-	-
		MPS (Quimb)		
Fourier*	-	-	-	-
Interp.*	-	-	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	-	-	-	-
TQA*	-	-	-	-
TQA	-	-	-	-
		PP		
Fourier*	-	17132.5 ± 0.0	-	-
Interp.*	-	16628.0 ± 0.0	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	-	-	-	-
TQA*	-	-	-	-
TQA	-	-	-	-
		SV		
Fourier*	4779.6 ± 762.1	2185.7 ± 0.0	9151.8 ± 3924.7	7659.2 ± 3043.9
Interp.*	4055.4 ± 575.4	1553.8 ± 0.0	3306.2 ± 2844.6	3000.7 ± 2768.1
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	4616.4 ± 710.9	2123.9 ± 0.0	9212.5 ± 4035.7	7871.9 ± 3175.0
TQA*	-	-	-	3448.1
TQA	-	-	-	3299.8

Table 56: **Avg. Energy $\pm$ standard deviation for MIS for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	R
MPS (Aer)								
Fourier*	-	17318.8 ± 27.4	-	-	-	10 [144]	-	
Interp.*	-	16162.6 ± 379.1	-	-	-	10 [144]	-	
Linear Ramp*	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS	-	-	-	-	-	-	-	
TQA*	-	-	-	-	-	-	-	
TQA	-	-	-	-	-	-	-	
MPS (Quimb)								
Fourier*	-	-	-	-	-	-	-	
Interp.*	-	-	-	-	-	-	-	
Linear Ramp*	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS	-	-	-	-	-	-	-	
TQA*	-	-	-	-	-	-	-	
TQA	-	-	-	-	-	-	-	
PP								
Fourier*	-	17132.5 ± 0.0	-	-	-	10 [144]	-	
Interp.*	-	16628.0 ± 0.0	-	-	-	10 [144]	-	
Linear Ramp*	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS	-	-	-	-	-	-	-	
TQA*	-	-	-	-	-	-	-	
TQA	-	-	-	-	-	-	-	
SV								
Fourier*	4779.6 ± 762.1	2185.7 ± 0.0	9151.8 ± 3924.7	7659.2 ± 3043.9	10 [20]	10 [21]	10 [20]	7
Interp.*	4055.4 ± 575.4	1553.8 ± 0.0	3306.2 ± 2844.6	3000.7 ± 2768.1	10 [20]	10 [21]	10 [20]	7
Linear Ramp*	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS	4616.4 ± 710.9	2123.9 ± 0.0	9212.5 ± 4035.7	7871.9 ± 3175.0	10 [20]	10 [21]	10 [20]	7
TQA*	-	-	-	3448.1	-	-	-	1
TQA	-	-	-	3299.8	-	-	-	1

Table 57: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

20 Tables for depth $P = 10$

	ER	HH	LB	RR
		MPS (Aer)		
Fourier*	-	10 [144]	-	-
Interp.*	-	10 [144]	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	-	-	-	-
TQA*	-	10 [144]	-	-
TQA	-	10 [144]	-	-
		MPS (Quimb)		
Fourier*	-	-	-	-
Interp.*	-	-	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	-	-	-	-
TQA*	-	10 [144]	-	-
TQA	-	10 [144]	-	-
		PP		
Fourier*	-	10 [144]	-	-
Interp.*	-	10 [144]	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	-	-	-	-
TQA*	-	10 [144]	-	-
TQA	-	10 [144]	-	-
		SV		
Fourier*	10 [20]	10 [21]	10 [20]	7 [20]
Interp.*	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	10 [20]	10 [21]	10 [20]	7 [20]
TQA*	-	-	-	1 [20]
TQA	-	-	-	1 [20]

Table 58: **Number of Instances (num. nodes in brackets) for** $P = 10$**.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
		MPS (Aer)		
Fourier*	-	17326.1 ± 29.2	-	-
Interp.*	-	16072.7 ± 324.4	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	-	-	-	-
TQA*	-	17298.2 ± 14.8	-	-
TQA	-	17167.3 ± 2.4	-	-
		MPS (Quimb)		
Fourier*	-	-	-	-
Interp.*	-	-	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	-	-	-	-
TQA*	-	17178.2 ± 11.7	-	-
TQA	-	16800.2 ± 1.5	-	-
		PP		
Fourier*	-	17128.7 ± 0.0	-	-
Interp.*	-	16575.6 ± 0.0	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	-	-	-	-
TQA*	-	17142.7 ± 0.0	-	-
TQA	-	16835.4 ± 0.0	-	-
		SV		
Fourier*	4800.7 ± 779.0	2182.8 ± 0.0	9305.2 ± 4020.2	7689.7 ± 3060.3
Interp.*	4051.0 ± 615.0	1607.1 ± 0.0	3504.3 ± 2465.6	2826.1 ± 2759.5
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS	4616.4 ± 710.9	2124.1 ± 0.0	9223.0 ± 4053.3	7873.1 ± 3175.7
TQA*	-	-	-	3470.8
TQA	-	-	-	3276.3

Table 59: **Avg. Energy ± standard deviation for MIS for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	R
MPS (Aer)								
Fourier*	-	17326.1 ± 29.2	-	-	-	10 [144]	-	
Interp.*	-	16072.7 ± 324.4	-	-	-	10 [144]	-	
Linear Ramp*	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS	-	-	-	-	-	-	-	
TQA*	-	17298.2 ± 14.8	-	-	-	10 [144]	-	
TQA	-	17167.3 ± 2.4	-	-	-	10 [144]	-	
MPS (Quimb)								
Fourier*	-	-	-	-	-	-	-	
Interp.*	-	-	-	-	-	-	-	
Linear Ramp*	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS	-	-	-	-	-	-	-	
TQA*	-	17178.2 ± 11.7	-	-	-	10 [144]	-	
TQA	-	16800.2 ± 1.5	-	-	-	10 [144]	-	
PP								
Fourier*	-	17128.7 ± 0.0	-	-	-	10 [144]	-	
Interp.*	-	16575.6 ± 0.0	-	-	-	10 [144]	-	
Linear Ramp*	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS	-	-	-	-	-	-	-	
TQA*	-	17142.7 ± 0.0	-	-	-	10 [144]	-	
TQA	-	16835.4 ± 0.0	-	-	-	10 [144]	-	
SV								
Fourier*	4800.7 ± 779.0	2182.8 ± 0.0	9305.2 ± 4020.2	7689.7 ± 3060.3	10 [20]	10 [21]	10 [20]	7
Interp.*	4051.0 ± 615.0	1607.1 ± 0.0	3504.3 ± 2465.6	2826.1 ± 2759.5	10 [20]	10 [21]	10 [20]	7
Linear Ramp*	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS	4616.4 ± 710.9	2124.1 ± 0.0	9223.0 ± 4053.3	7873.1 ± 3175.7	10 [20]	10 [21]	10 [20]	7
TQA*	-	-	-	3470.8	-	-	-	1
TQA	-	-	-	3276.3	-	-	-	1

Table 60: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.